

A Vision-Language Based Framework for Extracting and Understanding Classroom Content in Dual Languages

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing my/our degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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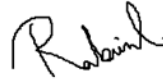
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Ethics Statement

The lecture recordings used in this research are recordings of the authors themselves. Three people appear as lecturers across the corpus, and all three are the members of this team. Nobody outside the team was recorded, asked to teach, or asked to take part, so the dataset contains no third-party participant and no consent was sought from anyone other than ourselves. The recordings were made in classrooms of Brac University and in other classrooms available to us.

No student was present as a subject of the recording. The camera faces the whiteboard and the person writing on it, and no other person’s face, name or voice forms part of the dataset used for training or testing. No personal information about any student was collected at any point.

The recordings, the transcripts and the reconstructed whiteboard images are kept in a private repository and are not published. The board images show the lecturer only as an occluding region that the pipeline removes, and the figures printed in this report were checked so that no person other than an author is identifiable in them. No lecture content is sent to an external service, because every model runs on our own machines.

Every transcript used as ground truth was produced and checked by members of the team, word by word against the audio. No number in this report was written down before the command that produces it was run, and the command for every number is recorded with the project so that any result can be reproduced. Results that did not support our expectations are reported in the same detail as the ones that did.

Abstract

University teaching in Bangladesh is usually delivered in a mixture of Bengali and English, a code-mixed form commonly called Banglish when it is written in the Roman alphabet. Existing speech recognition does not transcribe it. A multilingual model returns an English translation of what was said and falls into repetition loops on part of the material; a Bengali model returns Bengali script and spells the English technical terms phonetically. Neither preserves what the lecturer actually said, and a faithful transcript of such a class therefore has to be written in the Roman alphabet, which is the only way to carry both languages in one stream. The whiteboard, which holds most of the worked examples in a class taught without slides, is lost in both cases.

This research presents InsightLens, a four stage system that converts a recorded Banglish whiteboard lecture into a lecture note. Whisper large-v3-turbo adapted with Low-Rank Adaptation produces a romanised Banglish transcript. Each whiteboard is rebuilt from the video by taking every tile of the image from a moment when the lecturer was not standing in front of it, so no pixel is synthesised. A vision-language model then names and transcribes numbered regions of the rebuilt board, and a language model writes one section of notes per board with every quotation checked against the transcript.

A corpus of 44 lectures from three lecturers was recorded and 5.15 hours of it was transcribed by hand. On six held out lectures, fine-tuning reduces the median per-clip character error rate from 67.7 to 15.8 per cent across two random seeds. Changing the board reading prompt raises recall of hand-verified board content from 31.2 to 88.8 per cent, while a larger model is worth 5.3 points. The generated notes carry 89.1 per cent of the board content against 37.2 per cent for a conventional summarisation prompt, and in a blind pilot twenty readers preferred them to the earlier notes, 15 of the 19 who expressed a preference. Four approaches that failed are reported with the same detail, and the evaluation shows that measures built from English word lists systematically favour a model that translates code-mixed speech over one that transcribes it.

Keywords: Code-mixed speech recognition; Banglish; Bengali-English; Low-Rank Adaptation; Whisper; vision-language models; whiteboard reconstruction; lecture note generation

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Nomenclature

The next list describes several symbols & abbreviations that are used later within the body of the document

ASR Automatic Speech Recognition

Banglish Bengali-English code-mixed speech written in the Roman alphabet

CER Character Error Rate

LLM Large Language Model

LoRA Low-Rank Adaptation

LOSO Leave-One-Speaker-Out

OCR Optical Character Recognition

VLM Vision-Language Model

WER Word Error Rate

Chapter 1

Introduction

1.1 Background

Teaching in Bangladeshi universities is rarely monolingual. In a typical computer science class the lecturer explains an idea in Bangla, names it in English, writes the English term on the whiteboard and returns to Bangla for the worked example. A single sentence carries both languages, as in *Ekhon amra ei expression-er logical circuit dekhbo* , where the grammar is Bengali and the technical word is English. Written down, this mixture is normally rendered in Roman letters rather than in the Bengali script, and the result is commonly called Banglish. That is the form a transcript of such a class has to take. It is the only way to write both languages in a single stream without committing the sentence to one of them, so it is what a faithful record of the lecture looks like.

Automatic speech recognition performs well on the languages it was built for. Whisper is trained on roughly 680,000 hours of multilingual audio and reaches usable accuracy on many languages, including Bengali [1]. Such models assume, however, that an utterance belongs to one language. Whisper receives a language token and a task token, and on Bengali audio containing English words it usually produces a fluent English translation instead of a transcript. A Bengali model trained on clean Bengali writes everything in Bengali script and spells the English technical terms phonetically. Neither output preserves the lecturer’s words, and neither can be searched for the terms that were actually spoken.

Speech is also only half of a whiteboard lecture. The classrooms recorded for this work use no slides. Definitions, truth tables, SQL queries, subnetting calculations and diagrams appear on the board and are then discussed, so a note taken from the audio alone loses them. Recovering the board from the video is not a screenshot problem either. The lecturer stands in front of what he has just written, the ceiling lights leave glare on the surface, and the board is wiped and reused several times in one class.

Three developments make a system that handles both sides practical on ordinary hardware. Parameter efficient fine-tuning, in particular Low-Rank Adaptation [2], adapts a large speech model on a few hours of new speech by training a small number of additional weights, which fits in the memory of one desktop graphics card. Open vision-language models such as Qwen2.5-VL read text inside an image without a

dedicated optical character recognition stage [3]. Instruction tuned language models such as Qwen2.5-7B can take a transcript and board content together and write a structured document from them [4].

This research combines the three into a system called InsightLens. A lecture video is the input. The output is a lecture note containing the board redrawn without the lecturer in front of it, numbered coloured boxes around each part of the board, an explanation written from the lecturer’s own words, and quotations checked against the transcript. The note is produced in Banglish and in English so that the student can choose. Figure 1.1 shows the idea.

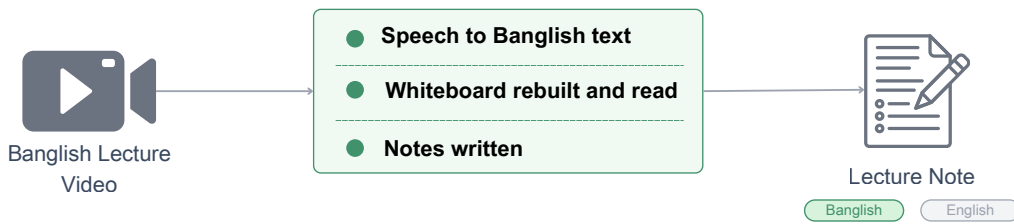


Figure 1.1: The InsightLens idea. A recorded Banglish lecture becomes a lecture note that contains both the speech and the whiteboard.

1.2 Rationale of the Study or Motivation

A student who misses a class, or who cannot follow the English terms at the speed they are spoken, currently has only the recording. Three hours of video is not a study aid. Passing that video through an available transcription service returns either English prose that was never spoken or Bengali script that nobody in the class writes, and neither contains the whiteboard.

The second motivation is cost. Producing a correct Banglish transcript by hand took about one hour of work for every ten minutes of video in this project, because existing tools give a transcriber nothing usable to start from. At that rate one course for one semester is several weeks of work. Bringing a model close enough to the lecturer’s words that a person only corrects small errors would reduce that cost substantially, which is why the error rate of the speech model is treated here as a result in its own right rather than as an internal detail.

The third motivation is that the gap is specific and checkable. Code switched speech recognition has been studied for Hindi-English and several other pairs, and Bengali speech recognition has its own literature. What is missing is the combination a Bangladeshi classroom needs, which is recognition of Bengali-English code-mixed

speech written back in the Roman alphabet and evaluated on real lectures and not on read or synthetic speech. Chapter 2 reviews the systems that come closest and states which part of this work is new.

The whiteboard side has value beyond note taking. A clean picture of every board a lecturer wrote, with the person removed and the content available as text, can be searched and reused, and it is a record of a class that was otherwise never written down.

1.3 Problem Statement

Given an ordinary recording of a university lecture delivered in Bengali-English code-mixed speech, in a room with a whiteboard and no slides, produce a lecture note that is faithful to what the lecturer said and to what the lecturer wrote.

Four smaller problems make this difficult, and each was confirmed by measurement on our own recordings before any solution was attempted. The first is that existing systems do not transcribe the speech. On the six lectures held out for testing, off-the-shelf Whisper large-v3-turbo produces a median character error rate of 67.7 per cent and a median word error rate of 93.9 per cent against human references. The failure is systematic, not random. The model translates into English and, on thirteen of the 177 test clips, repeats a phrase until it reaches the token limit. Table 1.1 shows one segment of a digital logic class in three versions.

Table 1.1: One segment of lecture BanglaASR11, as spoken and as produced by two models.

Source	Text
Human reference	<i>done. so x-or jodi amra bujhe jai tahole last je exclusive gate sheitao kintu amader bujha easier hoye jabe.</i>
Off-the-shelf	<i>so, x or gate and x or gate and x or gate and x or gate and ... (repeats to the token limit)</i>
Fine-tuned	<i>done. so, x or jodi amra bujhe jai, tahole last je exclusive gate, sheta o kintu amader bujha easier hoye chabe.</i>

The second is that the person who wrote on the board is standing in front of it. Measured over 79 frames of one lecture, the lecturer covers 23.5 per cent of the writing surface in the median frame and 7.0 per cent in the clearest one, and no frame is below 5 per cent. There is no moment at which a clean photograph of the board can simply be taken.

The third is that a summarising model writes from memory instead of from the lecture. The first version of this pipeline asked a language model for clear notes on the lecture, and the model produced textbook material, including a definition of a NAND gate that the lecturer never gave and that was wrong. Scored against hand-checked records of what was written on the board, those notes contained 37.2 per cent of the board items across 35 boards, and only six of the 67 numbers.

The last is that the obvious ways of scoring the system are misleading. A term based F1 over a fixed list of 82 English words, inherited from the earlier phase of this project, moves by 8.1 percentage points between two training runs on identical data, and it moves in the wrong direction. A model that transcribes the mixed speech correctly emits fewer English words than a model that translates it, so the better transcript scores worse. Any measure built from English strings behaves this way, and part of the contribution of this research is to document it.

1.4 Objective

The aim is to build and evaluate a system that converts a recorded Banglish whiteboard lecture into a usable lecture note, and to report how well each part of it works. The specific objectives are:

1. to build a hand-checked corpus of Bengali-English code-mixed classroom speech, transcribed in the Roman alphabet against a written standard, large enough to fine-tune and test a speech model;
2. to adapt a pretrained speech recognition model to this speech with Low-Rank Adaptation and measure the change in error rate on lectures never used for training;
3. to choose the training settings by a procedure written down before any result is seen and scored on a separate validation set, so that the reported test result is not the product of repeated tuning on the test data;
4. to rebuild each whiteboard from the video without the lecturer in front of it and without synthesising any pixel, and to measure how much of the writing survives;
5. to have a vision-language model read each rebuilt board and measure how much of its content is recovered against records checked by hand;
6. to generate a lecture note in Banglish and English that is grounded in the board and the transcript, points at numbered regions of the board image, and quotes the lecturer only in words that appear in the transcript;
7. to evaluate each stage with paired statistical tests and ablations, and to report the approaches that did not work as fully as the ones that did.

Four of the findings that came out of this work are not tied to the corpus and would hold for another language or another classroom. First, for reading a handwritten board with a vision-language model, the prompt matters about eleven times more than the size of the model: 57.6 points of board-content recall against 5.3. Second, generated notes can be scored on facts that were physically written on the board, which the note-writing model cannot supply from prior knowledge, and that separates grounding from general knowledge in a way fluency measures cannot. Third, measures built from English word lists are invalid for code-mixed speech, because they reward a system that translates and penalise one that transcribes correctly.

Fourth, no published Bengali or Bengali-English speech model produces romanised Banglish, which is shown here by running five of them on the same test clips. Chapter 6 states all of these again with the numbers behind them.

1.5 Methodology in Brief

The system runs as four stages. Each stage writes a file that the next stage reads, so any stage can be re-run or replaced on its own.

The first stage handles the speech. Audio is extracted from the video and cut into segments at the boundaries given by the transcription standard. Whisper large-v3-turbo is adapted with LoRA adapters on the training lectures, and the adapted model writes a timestamped Banglish transcript for a lecture it has not seen.

The second stage handles the board. Frames are taken at a fixed interval and the points at which the board is wiped are detected, which divides the lecture into eras with one board per era. Within an era the frame is cut into overlapping tiles and each tile is taken from a moment when the lecturer was not standing in front of that tile. The lecturer is located by a pretrained person segmentation network together with a shadow mask, so every pixel of the result is unmodified camera output.

The third stage reads the board. Our own code finds the blocks of writing, draws a numbered coloured box around each, and a vision-language model names and transcribes each box. This follows the Set-of-Mark style of prompting [5], in which the marks are drawn on the image and the model answers per mark.

The fourth stage writes the note. For each board a language model receives that board’s boxes, the board text and the part of the transcript belonging to that board, and returns one section of notes that refers to the boxes by number and colour. A checker then deletes any quotation absent from the transcript and any reference to a box that does not exist, and the sections are assembled into a page in Markdown and HTML.

Evaluation is done per stage. Speech is scored by per-clip median word and character error rate against human transcripts with paired Wilcoxon signed-rank and sign tests. The board reconstruction is scored by the fraction of tiles that found a clear view and by a human check of whether any writing was lost. Board reading and note quality are scored by recall of items from hand-verified records of each board’s content.

1.6 Scopes and Challenges

The dataset is 44 lecture recordings from three lecturers at one university, 8.33 hours of video, of which 28 lectures covering 5.15 hours carry hand-checked transcripts. The subjects are introductory programming, digital logic, databases and computer networking. Every recording is a whiteboard class with no slides, taken with a fixed camera. Notes are produced in Banglish and English. Notes in the Bengali script were considered and dropped, since they would be produced by translation alone and would add nothing the system itself contributes. The system is built for recorded

lectures, not live captioning.

Two limits are stated at the outset. The speech model is not accurate enough to be used without a human check, and Chapter 5 gives the remaining error rate. The measure used for the notes counts whether board content reached the page, which is a checkable property, but it says nothing about whether the surrounding explanation is well written or correct. A blind pilot with twenty readers, reported in Section 5.1.7, found the notes preferred overall and on layout against the earlier pipeline’s, but ease of reading was the one question of the four that did not reach significance, and no reader was asked to study from a note and then answer questions about the lecture, so how well the notes support learning is still untested.

Five difficulties shaped the work. Banglish has no agreed orthography, so a written transcription guide was produced and enforced by a validator, and a spelling tolerant version of the error rate was computed to separate spelling disagreement from genuine error. No public corpus of Bengali-English code-mixed classroom speech in Roman script exists, so the corpus had to be recorded and transcribed by hand, which was the largest cost in the project. All training ran on one desktop with a 12 GB graphics card and all vision-language work on a second with a 32 GB card, which put the largest vision-language models out of reach. The configuration chosen by the tuning procedure diverged on one of its two random seeds and produced a model worse than doing nothing. Finally, measures built from English strings reward a model that translates, which affects two separate results reported in Chapter 5.

1.7 Team Overview

The research was carried out by three students of the Department of Computer Science and Engineering, Brac University, supervised by Dr. Md. Golam Rabiul Alam and co-supervised by Md. Tanzim Reza.

The work fell into three areas. Data collection and transcription was the largest shared task, since every transcript was checked word by word against the audio by a member of the team. Model training and evaluation covered the speech experiments, the tuning procedure and the statistical testing. The vision and note generation area covered board reconstruction, the vision-language prompting and the note builder. The board records used for scoring were checked by one member, reading each board image without seeing any model output, and all three members contributed to this report.

Chapter 2

Literature Review

2.1 Preliminaries

2.1.1 Code-mixing and Banglish

Code-switching is the use of more than one language by the same speaker in the same conversation. Two cases are usually distinguished. Inter-sentential switching means the speaker finishes a sentence in one language and begins the next in another. Intra-sentential switching, also called code-mixing, means the switch occurs inside a single sentence. The second case is harder for speech systems, because the utterance cannot be assigned to one language and handed to one monolingual model [6].

Bangladeshi classroom speech is strongly intra-sentential. In the recordings used here a single clause regularly carries a Bengali verb, an English noun phrase and an English technical term, so the speech is Bengali in grammar and often English in content words.

The written form matters as much as the spoken one. Bengali has its own script, but it is also widely written in the Roman alphabet in informal digital text, a form usually called Banglish. There is no spelling standard, so the same word appears as *ami*, *aami* or *amee* depending on the writer. Romanised Bangla is now an object of study in natural language processing in its own right. BanglaTLit, for example, is a benchmark of 42,700 romanised Bangla samples with a pre-training corpus of 245,700 more, built for converting romanised Bangla back into the Bengali script [7]. Work of this kind establishes romanised Bengali as a real written form with real users, which is the reason it was chosen here as the output of the speech model rather than the Bengali script.

2.1.2 Speech recognition, error rates and the models used

Automatic speech recognition takes an audio signal and returns text. Its usual measure is the word error rate, the number of word substitutions, insertions and deletions needed to turn the output into the reference divided by the number of reference words. Character error rate is the same calculation over characters. Both can exceed 100 per cent when the system produces more text than was spoken.

Two properties of these measures shaped how results are reported in this research. A single long wrong output for one short clip pushes a mean error rate above 100 per cent and hides everything else, so medians over clips are reported and systems are compared clip by clip. Word error rate is also unkind to a language with no spelling standard, because two acceptable spellings of the same word count as an error; Section 2.2.4 reviews what the code-switching literature does about this.

Modern recognisers are built in one of two ways, and the difference matters later in this research because the two write their output differently. The first is connectionist temporal classification, which scores an alignment between audio frames and a fixed set of output symbols [8]. A model of this kind can only ever emit characters that are in its output alphabet, which is decided before training. The second is the encoder-decoder transformer [9], which generates text one token at a time and can in principle produce any string its tokeniser can represent. Conformer models add convolution to the transformer encoder and are a common choice for speech [10].

Most recent recognisers for languages with little transcribed audio start from a model pretrained on unlabelled speech. wav2vec 2.0 learns from masked audio with a contrastive objective [11], HuBERT predicts hidden units discovered by clustering [12], and WavLM extends the idea to a wider set of speech tasks [13]; a review covers the family and how the approaches relate [14]. The same recipe has been scaled across many languages at once, first with XLSR [15] and then with XLS-R [16], which is where most published Bengali recognisers begin. WavLM is used directly in this research, not as a recogniser but to produce a voice embedding for each recording, which is how the lecturer of every video was confirmed. Where labelled audio is scarce, SpecAugment masks bands of time and frequency during training and is close to standard practice [17].

Whisper is a family of encoder-decoder transformer models trained on about 680,000 hours of weakly supervised multilingual audio [1]. Its decoder receives special tokens stating the language and the task, where the task is either transcribe or translate, which explains two behaviours central to this work. Because the model is told which single language it is hearing, mixed language audio is pushed towards one of them, and for Bengali audio containing English words the output is usually English. Because the decoder is autoregressive with a strong internal language model, it can enter a repetition loop in which the same phrase is emitted until the token limit is reached. The Whisper paper proposes a fallback rule for this, in which a hypothesis whose gzip compression ratio exceeds a threshold is decoded again with sampling [1], and WhisperX avoids the problem by segmenting audio with an external voice activity detector instead of relying on the model’s own timestamps [18]. Loops are reported in this research as a counted quantity.

A vision-language model takes an image and a text instruction and returns text. Open models such as Qwen2.5-VL read text inside an image well enough that a separate optical character recognition stage is often unnecessary [3]. Directing such a model at a specific part of an image is a separate problem, which Set-of-Mark prompting solves by drawing marks such as numbered boxes onto the image and asking the model to answer per mark [5]. The board reader in this research uses the same idea, with the boxes found by our own image processing and the numbers carried through into the generated notes.

An instruction tuned language model such as Qwen2.5-7B-Instruct can be given a transcript and asked for structured notes [4]. The risk is known from the summarisation literature. Models produce fluent text that the input does not support, and a study of state of the art abstractive summarisers found that a large share of generated summaries contain hallucinated content [19]. Faithfulness asks whether the output follows the input given to the model; factuality asks whether it is true of the world. A lecture note must be faithful, because a statement that is true in general but was not part of the lecture is still the wrong note.

2.1.3 Fine-tuning, validation, ablation and statistical testing

Fine-tuning means continuing to train a pretrained model on new data so that it fits a new domain. Training all weights of a model with hundreds of millions of parameters requires a large amount of memory, because the optimiser keeps extra state for every parameter, and it risks destroying what the model already knows when the new dataset is small.

A family of methods answers this by training a small number of new parameters and leaving the pretrained weights alone. Adapters insert small trainable layers between the frozen ones [20], and prefix tuning learns a short sequence of vectors that is prepended to the input instead of changing any weight [21]. Low-Rank Adaptation is the member of that family used here. A later variant, QLoRA, quantises the frozen model to four bits so that a much larger model fits on one card [22]; this research did not need it for the recogniser, but the same idea is what makes the note-writing model practical on a single consumer card.

Low-Rank Adaptation avoids both problems [2]. The original weight matrix is frozen, and beside it two small matrices are trained whose product has the same shape as the original, so the layer output becomes the sum of the frozen path and the small trained path. If the original matrix is $d \times k$ and the two small matrices have inner dimension r , the number of trained parameters falls from $d \times k$ to $r(d+k)$, and r is typically 8, 16 or 32. The frozen model is untouched, so the adapter can be removed or swapped. In speech work the adapters are usually placed on the attention projections, most often the query and value projections [23]. The rank r controls how much the adapter can change the model and the learning rate controls the size of each training step; both are hyperparameters, chosen by the person training the model rather than learned from the data.

Four further terms recur in Chapters 4 and 5. The training set is what the model learns from, the validation set is used to choose settings such as the learning rate, and the test set is used once at the end to report the result. Choosing settings by looking at the test set makes the reported number optimistic, because the choice has been fitted to data that is supposed to be unseen. Hyperparameter tuning is the search over settings, and it is defensible only when scored on validation data, with the order of the search and the rule for accepting a winner written down in advance. An ablation removes or changes one component and measures the effect, which is how a claim that a particular part is responsible for a result can be supported. Finally, in speech work it matters greatly whether the test speaker was heard during training,

so this research reports both an unseen speaker design and an unseen recording design and labels each one.

Training in this research uses AdamW, which decouples weight decay from the gradient update and is the usual optimiser for transformer fine-tuning [24], through the Hugging Face Transformers library [25].

Comparisons here are paired, because both systems are run on the same clips or boards. The Wilcoxon signed-rank test ranks the differences by size and asks whether positive and negative differences are balanced [26], while the sign test counts only how many items improved and how many got worse. Both are used, and the direction of the difference is reported beside every p-value.

2.2 Review of Existing Research

The work this research builds on falls into four areas. The first is speech recognition for Bengali and for code-switched speech, which supplies both the models and the evidence that code-switching is where they fail. The second is the small set of systems that target Bengali-English speech directly, which is where this research has to state what is new. The third is how code-switched recognition is measured, which matters here because two of the results in Chapter 5 are about a metric misleading rather than about a model improving. The fourth is reading a whiteboard and generating lecture notes, which is the vision half of the pipeline. Each is reviewed in turn, and the section ends with the gap the four leave open.

2.2.1 Bengali and code-switched speech recognition

Bengali speech recognition is an active area with substantial resources behind it. OOD-Speech is the largest public Bengali corpus of its kind, with 1177.94 hours of crowdsourced training audio from 22,645 native speakers and a 23.03 hour test set annotated by hand from seventeen sources, including television drama, audiobooks, talk shows and online classes [27]. Its purpose is to measure how badly models degrade when the test domain differs from the training domain, which is exactly the failure this research would meet by training on read speech and testing on a classroom. Two other resources sit behind most published Bengali systems. Common Voice is a crowdsourced multilingual corpus whose Bengali portion is the training set for a large share of the public models [28], and Multilingual LibriSpeech supplies read speech for the same purpose in other languages [29]. For Indian languages specifically, IndicSUPERB provides the Kathbath corpus and a benchmark across twelve languages, which is the closest thing the region has to a standard evaluation [30].

On the modelling side, applying wav2vec 2.0 to the Bengali Common Voice dataset gives a word error rate of 0.2524 on a validation split of 7,747 samples [31], and several Whisper models fine-tuned on Bengali Common Voice are publicly available. The Bengali branch of the earlier phase of this project used one of them, a Whisper-small model fine-tuned on Bengali Common Voice.

All of these write Bengali script. For a classroom whose technical vocabulary is

English that is a problem, not a solution, because the output turns *NAND gate* into a Bengali spelling of the English sound, which a search for the English term cannot match and which no longer records the word the lecturer actually used. This is not left as an argument: Section 2.2.3 reports what five of these published models actually produce when they are run on the test clips of this research.

Writing a South Asian language in the Latin alphabet is itself a studied problem. The Dakshina dataset pairs native-script and romanised text for twelve South Asian languages and documents how unstandardised the romanised side is [32], and Aksharantar is a large transliteration corpus built to train models that move between the two [33]. For Bengali specifically, BanglaTLit addresses the reverse direction, turning romanised Bangla back into Bengali script [7]. What none of this work does is treat the romanised form as the transcription target for speech, which is the position taken here. The reason is not that the romanised form is preferred for reading, but that it is the only target that records a code-mixed utterance without committing it to one language or the other.

Code-switched recognition has its own literature, surveyed in general terms by Sitaram et al. [34] and, for end-to-end systems, in a recent systematic review that catalogues the languages, datasets, metrics and model choices used across the field [6]. Two observations from that review apply here: intra-sentential switching is the hard case, and the field is short of natural, spontaneous, in-domain speech data, because most corpora are read or scripted. The best known exception is SEAME, about 192 hours of spontaneous Mandarin-English conversation, which has served as the standard benchmark for the field for over a decade [35]. Nothing of that kind exists for Bengali-English. Winata et al. also show that a multilingual model is not automatically a code-switching model, since handling two languages separately is not the same as handling them in one sentence [36].

For South Asian languages, Chadha et al. built code-switched recognition for Indic languages using wav2vec 2.0 with language identification, reporting word error rates of 21.77 for Hindi-English and 28.27 for Bengali-English [37]. Those figures are much lower than the ones reported here, but not because their system is better at this task. Their transcription target is the native script for the Bengali parts, their audio comes from a different and generally cleaner domain, and their test split is not made of whole unseen lectures. Error rates across different datasets and transcription targets are not comparable, and are not presented as such in this research.

On the efficiency side, Yang et al. study parameter efficient adaptation of Whisper for code-switching, comparing soft prompt tuning variants against LoRA on the SEAME and ASRU2019 Mandarin-English datasets, and find deep prompt tuning competitive with LoRA while preserving performance on the original languages [23]. Their conclusion supports the design used here, in which a small adapter on a frozen multilingual model adds a code-switched variety without retraining everything.

2.2.2 The nearest Bengali-English systems

Four efforts are close enough to this research to be named explicitly, and Table 2.1 compares them.

MediBeng is a translation system, not a transcription system, and its training data

Table 2.1: Systems that recognise Bengali-English code-switched speech, with the system built in this research. The error rates come from different datasets with different transcription targets and are not comparable with one another.

System	Model	Data	Output	Reported error
MediBeng Whisper Tiny [38]	Whisper-tiny, full fine-tune	Synthetic clinical dialogue	English translation	WER 0.01, BLEU 0.98
BEHE-CMDisfl [39]	Kaldi GMM-HMM baseline	Synthetic speech from LLM text and TTS, 1.3 h subset	Not stated	WER 37.74%
Indic code-switched ASR [37]	wav2vec 2.0 with language ID	Indic code-switched corpora	Native script	WER 28.27 (Bengali-English)
Whisper-small-benglish (public model, no paper)	Whisper-small fine-tune	194 clips of about 30 s, roughly 1.6 h	Not documented	WER 0.4123 on a random 10 per cent split
This research	Whisper-large-v3-turbo with LoRA	5.15 h of hand-checked classroom speech, 28 lectures, 3 lecturers	Romanised Banglish	CER 67.7 to 15.8, WER 93.9 to 41.7, six whole unseen lectures

is synthetic [38]; a word error rate of 0.01 with a BLEU score of 0.98 is what synthetic data produces and should not be read as accuracy on real clinical speech. BEHE-CMDisfl is likewise synthetic, generated by prompting a language model for code-mixed text with disfluencies and rendering it with text-to-speech [39]; it is a useful resource and makes the same point about missing spontaneous data, but it is not spontaneous speech. The Indic work uses real speech but targets the native script [37]. The community model Whisper-small-benglish is closest in intent, since its card names lecture notes among its uses, but it is a model card and not a study. It is trained on about 1.6 hours and evaluated on a random ten per cent of the same clips, so the same speakers and very likely the same recordings appear on both sides of the split, and no example output is given, so its orthography cannot be confirmed. A public repository by Asrarul applies LoRA to Whisper-large-v3 for Bengali-English meeting speech and is the closest match in method, but reports no results.

The conclusion is narrower than a claim of the first Banglish speech recogniser. Bengali-English code-switched recognition exists. What could not be found is a system that transcribes spontaneous Bengali-English classroom speech into romanised Banglish, evaluated on whole lectures held out from training against hand-checked references, with the whiteboard read alongside it. Every claim of novelty in this research is written in that form.

2.2.3 What the published models produce on our data

The error rates in Table 2.1 come from different datasets with different transcription targets, so they cannot be compared with one another or with this work. An argument of that kind is weaker than a measurement, so five published models were run on the 177 held-out clips of this research, with the same references, the same normaliser and the same metric code that produced the result in Chapter 5. Table 2.2 gives the outcome.

Two decisions keep the comparison fair. Each model decodes under its own generation settings, with no language or task token imposed on it, because a Bengali model was trained to emit Bengali and forcing it into another mode would break it for reasons that would be our fault. And because our references are romanised, a model that answers in Bengali script scores badly no matter how well it heard the speech, so the alphabet each model wrote in is counted and reported beside its error rate.

Table 2.2: Published Bengali and Bengali-English models run on the 177 held-out clips of this research. Median per-clip rates. The transliterated column romanises each model’s Bengali output before scoring and keeps the better of two schemes per clip, so it is an upper bound on what the model could score if orthography were free.

Model		CER	CER after translit.	WER	What it wrote
tugstugi	Bengali	100.0	47.1	100.0	Bengali script, all 177 clips
Whisper-medium					
Whisper-small-benglish		89.9	66.0	95.8	Bengali script 124, mixed 53
bangla-ASR-v5		98.6	73.0	100.0	Bengali script, all 177 clips
BanglaASR		98.6	73.8	100.0	Bengali script, all 177 clips
MediBeng Whisper-tiny		109.2	n/a	146.5	English translation, 70 runaways
Whisper-large-v3-turbo, off the shelf		67.7	n/a	93.9	English translation
This research		15.8	n/a	42.5	Romanised Benglish

Not one of the five produced romanised Banglish on a single clip. Four write Bengali script and the fifth translates into English. The clearest case is the model whose card advertises Benglish, which does switch between languages but writes the Bengali half in Bengali letters: where the lecturer said *hello everyone, welcome to the second class of digital logic design*, it returned *Hello* followed by the English word *everyone* spelled out in Bengali script, then the English clause, then Bengali script again. The strongest Bengali model in the table writes Bengali script so consistently that none of its output survives conversion to plain ASCII at all, which is why both of

its error rates are exactly 100.

The obvious objection is that this punishes a model for its alphabet rather than for its hearing, so the Bengali output was transliterated to Roman and scored again. That closes much of the gap and it must be reported honestly. The strongest model improves from 100.0 to 47.1, which is better than off-the-shelf Whisper, so it would be wrong to say these models are no better than an untuned system. It does not close the comparison: 47.1 against 15.8 is still three times the error, and word error rate barely moves at all, from 100.0 to 90.2, because transliteration can fix an alphabet but cannot invent the spelling of a language that has no spelling standard.

The conclusion is therefore narrow and measured. These are capable models being asked for something they were not built to produce. No published system was found that outputs romanised Banglish, and the transcripts they do produce cannot be read by the student this research is written for. One candidate, a wav2vec 2.0 Bengali model, was not run; its output alphabet was inspected instead and is dominated by Bengali characters, but it is not claimed here as measured.

2.2.4 Measuring code-switched recognition

Evaluation is a recognised research topic in this area, and it bears directly on two results reported in Chapter 5. Hamed et al. benchmarked evaluation metrics for code-switching recognition against human judgements and found that transliteration followed by text normalisation agrees most closely with human acceptance [40]. Emond et al. proposed a transliteration based word error rate for the same reason, mapping mixed script output into a single script before scoring [41]. PolyWER accepts several valid forms of a reference, including transliterations and translations, on the grounds that plain word error rate is too strict for speech that can be correctly written in more than one way [42].

The common thread is that a measure built around one script or one vocabulary misjudges a system that legitimately produces another. This research reaches the same conclusion from a different direction. Its inherited term based measure counts a fixed list of English words, so a model that transcribes the mixed speech faithfully emits fewer of them and scores worse for being right. Section 5.4.2 quantifies the effect.

2.2.5 Whiteboard content and lecture note generation

Extracting what is written on a board from a fixed camera lecture video is an established line of work in document analysis. Yeh et al. extract handwriting and produce a video summary by statistical modelling and clustering of board pixels, robust to noise, lighting change and occlusion [43]. Davila and Zanibbi build a spatio-temporal index of content regions and segment the video by detecting conflicts between them, producing static image summaries of a whole lecture from a minimal set of frames with 96.28 per cent recall of connected components [44]. Kota et al. generalise this with a deep network that detects handwritten text, formulae and sketches and then binarises them, reaching an f-measure of 94.32 per cent for binary connected components on the AccessMath dataset [45].

Reading text out of an image has itself changed since that work was published. Document understanding first moved to models that combine text, layout and image features, such as LayoutLM [46], and then to models that skip optical character recognition altogether and map the image straight to the output string, such as Donut [47]. General vision-language models followed. Visual instruction tuning produced open models that answer questions about an image in natural language [48], closed models of the same kind read text in photographs well [49], and a recent survey covers how these models are now applied to visually rich documents and where they still fail [50]. A whiteboard is a harder case than a scanned page, because the writing is handwritten, unevenly lit, partly occluded by the person writing it, and never captured straight on.

These systems solve a closely related problem and solve it well. Two differences matter here. Their output is a binarised summary image or an index of handwriting regions, which is the right target for search and navigation, whereas the target in this research is a photograph-like board that a student can look at inside a note, with the original strokes and colours intact. None of them reads the content into text with a model that understands what it is reading, because that capability did not exist when most of them were published. The board method used here is a tiled mosaic over erase separated eras, which is engineering built from known parts and not a new algorithm, and Chapter 4 describes it as such.

Turning a recorded lecture into notes is an old goal given new life by large models. M3AV is a recent academic lecture dataset of nearly 367 hours across five sources with human annotation of slide text and spoken words, built to support contextual speech recognition, slide generation and script generation [51]. NoteIt converts instructional videos into interactable notes by extracting hierarchical structure and multimodal key information, and evaluates the result both on technical metrics and with a user study of 36 participants [52]. Two gaps appear when this work is placed beside the classroom targeted here. Existing systems assume slides, which are text that can be read reliably or taken directly from a file, whereas our lecturers write on a whiteboard and stand in front of what they write. The datasets and systems are also built for English or Chinese and for speech that stays in one language. The user study in NoteIt points at something this research has not done. Whether notes are good is a question about readers, and it is answered by asking readers.

Results from vision-language models in this area are also strongly affected by the prompt. Promptception, a systematic study of prompt sensitivity across ten large multimodal models, reports accuracy swings of up to 15 per cent from minor changes in phrasing and structure [53]. That finding is the published context for one of the results in Chapter 5, in which changing the board reading prompt was worth far more than changing the size of the model.

2.3 Summary of Key Findings

The review supports four conclusions. The first is that the speech problem is real and is not solved by anything available. Bengali speech recognition is mature but writes a script that discards the code-mixing a faithful transcript has to keep. Code-switched recognition for Bengali and English exists, but the published systems either

translate into English, train on synthetic speech, or target the native script, and the one community model that matches the intent most closely is undocumented and evaluated on a random split of its own clips.

The second is that the board problem is well studied, and this research should not claim otherwise. Extracting handwritten content from fixed camera lecture video has a long literature with strong results on public datasets. What is new here is not the extraction but what follows it, which is reading the rebuilt board with a vision-language model and carrying numbered regions of it into a generated note.

The third is that measurement is an open problem in this area and not a detail. Several independent groups have found that standard word error rate misjudges code-switched output and have proposed transliteration based or multi-reference alternatives. The same class of problem appears twice in this research, once in the inherited term measure and once in the note quality measure.

The last is that lecture note generation with large models is active but assumes a different classroom. Existing systems expect slides and a single language, and nothing found combines code-mixed speech, a handwritten whiteboard and generated notes in two languages. Table 2.3 states the gap in the form the rest of the research uses.

Table 2.3: The gap this research addresses.

Component	What already exists	What this research adds
Speech	Bengali ASR in native script; Bengali-English code-switched systems on synthetic or out of domain data	Romanised Banglish transcription of spontaneous classroom speech, tested on whole unseen lectures against hand-checked references
Evaluation	Transliteration based and multi-reference metrics for code-switched ASR	A measured demonstration, in two places, that English-string measures reward a translating model and penalise a transcribing one
Whiteboard	Handwriting extraction, binarisation and summary images from fixed camera video	A reconstruction in real camera pixels, read as text by a vision-language model and scored against hand-verified records
Notes	Slide based lecture summarisation and note generation in English	Notes in two languages that point at numbered regions of the board, with quotations checked word for word against the transcript

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

3.1.1 Functional and non-functional requirements

Table 3.1 lists what the system must do. Each requirement has a short identifier so that later chapters can refer to it.

Seven non-functional requirements constrain how those functions are delivered. The system runs on one consumer graphics card, with speech training fitting in 12 GB using gradient checkpointing and the vision-language work in 32 GB, and uses no cluster and no paid cloud service. It is reproducible. Every reported number has a command that regenerates it, and splits, seeds and chosen settings are written to disk beside the model. It is restartable, because each stage writes its output to a file and is skipped when that file exists. Nothing in the board image is fabricated, since every pixel comes from a real frame and generative inpainting is not used. Claims in the note are traceable, because quotations are checked against the transcript, board content is supplied as text and non lecture material is labelled. The system runs offline once the model weights are downloaded, so no lecture audio, video or text is sent to an external service. Finally, recordings and transcripts are held in a private repository and the public output of the project contains no raw recording.

3.1.2 Data requirements

Three kinds of data were needed, and the cost of producing them shaped the schedule. The first is the recordings themselves. Forty-four classroom recordings were collected from three lecturers, 8.33 hours in total, each a fixed camera view of a lecturer at a whiteboard with no slides. The subjects are introductory programming, digital logic, databases and computer networking.

The second is the reference transcripts. Twenty-eight of these lectures, covering 5.15 hours of speech in 609 timed segments and about 39,100 words, were transcribed by hand. Every transcript follows the written standard of Appendix A and is validated

Table 3.1: Functional requirements.

ID	Requirement	Where met
FR1	Accept a recorded lecture video and extract a 16 kHz mono audio track.	Section 4.3
FR2	Produce a timestamped transcript in romanised Banglish using a model adapted to this speech.	Section 4.4.1
FR3	Detect the points at which the board is wiped and treat the writing between two wipes as one board.	Section 4.4.2
FR4	Rebuild each board without the lecturer in front of it, using only pixels that appear in some real frame.	Section 4.4.2
FR5	Find the blocks of writing on a rebuilt board and draw a numbered coloured box around each one.	Section 4.4.3
FR6	Name and transcribe the content of each numbered box with a vision-language model.	Section 4.4.3
FR7	Generate a lecture note, one section per board, referring to the boxes by number and colour.	Section 4.4.4
FR8	Produce the note in Banglish and in English, quoting the lecturer verbatim in the Banglish version and in translation in the English one.	Section 4.4.4
FR9	Delete any quotation absent from the transcript and any reference to a box that does not exist.	Section 4.4.4
FR10	Keep material the lecturer did not say inside a separate labelled section.	Section 4.4.4
FR11	Output the note as Markdown and as one self-contained HTML page with the board images embedded.	Section 4.4.4
FR12	Record how long each stage took and how many quotations were kept and deleted.	Section 4.4.5

by a checking script. The remaining sixteen recordings have no transcript and support the vision side and the demonstration only.

The third is the board records. To measure whether board content reaches the notes, the contents of 45 boards were written out by hand as lists of items, 436 items in total, each labelled as a number, name, code fragment, term or phrase. These records are used only for scoring and are never given to a model. A separate human check covered a further 98 boards and recorded whether anything written was missing from the reconstruction.

3.1.3 Hardware, software and constraints

All speech work ran on a desktop with an RTX 3060 of 12 GB, covering data preparation, hyperparameter tuning, LoRA training, evaluation, board reconstruction and the person segmentation network. All Qwen work ran on a second desktop with an RTX 5090 of 32 GB, covering board reading, box naming and note generation for every lecture. The first version of the pipeline, whose results are reported separately where they are still used, ran on a third machine with an RTX 3090 of 24 GB.

The software is Python 3.11 with PyTorch 2.5.1 for CUDA 12.1 on the first desktop

and Python 3.12 with PyTorch 2.11 for CUDA 12.8 on the second, which the newer card requires. Both run Transformers 5.12.1 with PEFT for the LoRA adapters, and FFmpeg extracts audio and frames. The models are whisper-large-v3-turbo for speech, Qwen2.5-VL-7B-Instruct and Qwen2.5-VL-3B-Instruct for board reading, Qwen2.5-7B-Instruct for note writing, a pretrained DeepLabV3 with a ResNet-101 backbone for person segmentation [54], and a WavLM speaker embedding model used only to confirm which lecturer appears in which recording. About 31 GB of model weights and 75 GB of video have to be stored.

Four constraints were fixed. The corpus is what three lecturers agreed to record and what the team could transcribe by hand in the time available, which bounds how strong any speech result can be. Training the 809 million parameter speech model on a 12 GB card is possible only with gradient checkpointing, and the largest vision-language models were out of reach, since Qwen2.5-VL-32B is 68.3 GB of weights and the 72B version 146.8 GB. The project runs to a fixed submission date, so a fourteen hour run can be started once instead of five times. The fourth constraint was self-imposed and removed the most work: no number enters this report unless a command in the project regenerates it. An earlier version of the project carried figures whose numbers had been typed instead of computed, and those were removed.

3.2 Societal Impact

The immediate beneficiaries are students in classrooms where teaching is code-mixed. A student who missed a class, who is slower at English, or who wants to revise currently has a video and nothing else, and the system turns that video into a document containing the board. Because the note exists in Banglish as well as English, a student who thinks in the mixed language is not forced to read a translation of their own lecture. Lecturers also gain a record of every board they wrote, which nothing else in a whiteboard class preserves and which costs them no extra work.

Language technology serves widely spoken standard languages well and mixed speech badly, so a student whose classroom language is Bengali-English code-mixed is poorly served by both English tools and Bengali tools. Treating that mixture as a first class target, with its own transcription standard and its own written form, is a step against that imbalance, and the same argument applies to the choice of the Roman alphabet. A text transcript and a text version of the board also make a lecture reachable by a screen reader and searchable by keyword, which a video is not, although the system was not designed as an assistive technology and has not been tested with users who have a disability.

Three risks deserve naming. Generated notes are not guaranteed to be correct, and Chapter 5 gives a real example in which the model wrote that the two universal gates are XOR gates when the lecturer and the board both said NAND and NOR. A student who trusts such a note will learn the wrong thing, so any deployment must present the notes as machine generated study aids to be checked against the lecture, and the system keeps the board image beside the text so that a reader can check. A lecture recording also captures a person at work and may capture student voices; this project used recordings made with the lecturers' permission and kept them private,

and any wider use needs consent from the people recorded. Finally, good automatic notes may reduce attendance, and taking notes by hand has a learning value of its own. That trade-off is not resolved here, and the system is aimed at revision and at students who could not attend, not at replacing attendance.

There is no physical safety dimension to the work. Lecture recordings and white-board content belong to the lecturer and the university, so recordings are not redistributed and released model weights are used under their own licences. Culturally, the system preserves the lecturer’s own mixed speech instead of converting it into standard English, which respects how the class is taught.

3.3 Environmental Impact

Training machine learning models consumes electricity, and the field has been criticised for not reporting it [55]. The headline figures in that literature come from very large models trained on clusters. The work in this research sits at the other end of the scale, and the actual figure is below.

All training is parameter efficient, so no full model was trained from scratch or fine-tuned end to end. Table 3.2 lists the machine time taken by the jobs that produced the results in Chapter 5, read from the job logs.

Table 3.2: Machine time for the jobs behind the reported results, from the job logs. The rows marked approximate are added from run records rather than a single log line.

Job	Machine	Time
Hyperparameter tuning, rehearsal, 7 trainings and evaluations	RTX 3060	about 5.4 h
Hyperparameter tuning on the full corpus, 7 trainings and evaluations	RTX 3060	6.0 h
Final model, seed 42	RTX 3060	1.3 h
Final model, seed 1	RTX 3060	1.1 h
Second learning rate, two seeds	RTX 3060	about 2.4 h
Transcripts and board reconstruction for all lectures	RTX 3060	about 2 h
Board reading, box naming and note generation, all lectures	RTX 5090	about 6 h
Total		about 24 h

The energy figure below is an estimate from the measured hours and stated assumptions and not a meter reading. Assuming a whole system draw of about 250 W for the first desktop while training and about 600 W for the second, the 18 hours on the first machine and 6 hours on the second come to roughly 4.5 and 3.6 kilowatt hours, so about 8 kWh in total. At a grid emission factor of about 0.65 kilograms of carbon dioxide equivalent per kilowatt hour, a reasonable figure for the Bangladesh grid, that is roughly 5 kilograms of carbon dioxide equivalent for the entire experimental programme. The widely cited estimate for training a single large language model with neural architecture search is over 280,000 kilograms [55].

Three decisions kept the figure small, and each was taken for other reasons as well. Using LoRA instead of full fine-tuning means a training run updates a few million parameters instead of 809 million, so a run finishes in about an hour on a mid-range card. Choosing hyperparameters by a plan with a fixed number of stages meant seven training runs instead of an open ended search. Not running the 32 billion and 72 billion parameter vision-language models avoided both a 68 GB download and the compute to use them, and the comparison in Section 5.2.2 shows the larger model was not where the important difference lay.

In use the system runs on one desktop, and processing a lecture end to end takes a few minutes of graphics card time, so a department could run it on hardware it already owns. There is no requirement for a data centre and no per-lecture payment to an external service, which also means no lecture content leaves the institution.

3.4 Ethical Issues

The people recorded are the authors. Three of us appear as lecturers across the corpus, and no one outside the team was recorded or asked to take part, so there is no third-party participant in the dataset and no external consent to obtain. The recordings were made in classrooms of Brac University and in other classrooms available to us, with no student present as a subject. The recordings are not published and are not shared outside the team. Where a board image appears in this report, the lecturer has been removed by the reconstruction itself, so the figures show writing rather than people. Videos, audio, transcripts and board images are kept in a private repository, the transcripts contain classroom speech only, and no student personal data was collected. No lecture content is sent to an external service at any point, because every model runs locally, which was one reason for preferring open model weights over a commercial interface.

The most serious professional risk in a project of this kind is reporting a number that was never computed, and this project had that problem in its earlier phase. A significance value, an effect size, an ablation table and a set of per-video statistics had been written into figures and text without any code behind them. They were found during a review, they are listed in the project record as retired claims, and none of them appears in this report. Two rules were adopted afterwards and kept. A number may be written only if a command in the project regenerates it, and results that go against the project's own hypothesis are reported in the same detail as results that support it. Chapter 5 contains several of the second kind.

A note generator can also state things the lecturer never said. The design contains three controls for this, described in Section 4.4.4, and Section 5.3 reports how well each one held, including the counted number of quotations that had to be deleted and the one box reference that survived the checker. The controls reduce the risk without removing it. There is an academic integrity dimension as well. Notes produced by a machine from a lecture are a study aid, and if a student submits them as their own writing that is plagiarism in the ordinary sense, which is one reason generated material inside the notes is labelled.

3.5 Standards

The project follows the standards and conventions in Table 3.3. Several are conventions of the field and not formal standards, and the table says which.

Table 3.3: Standards and conventions followed.

Area	Standard	How it is applied
Transcription	Project transcription guide, version 1.0	Segments of 10 to 25 seconds and never over 30, timestamps as [M:SS-M:SS], fillers kept, ASCII only, a fixed spelling table. A validator checks every file.
Audio	16 kHz mono PCM WAV	The input format the speech model expects.
Speech evaluation	Word and character error rate	Reported as per-clip medians with paired rank tests.
Interchange	UTF-8, JSON, Markdown, HTML5	Machine-readable results in JSON, notes in Markdown, and a self-contained HTML page with images embedded.
Code	PEP 8, Git	One script per stage, no deletion of experimental scripts, every result traceable to a commit.
Referencing	IEEE style	As required by the department template.
Licensing	Licences of the released models	Whisper, Qwen and the segmentation network are used under their publishers' terms and are cited.

3.6 Project Management Plan

The work ran in five phases over thirteen months, from September 2025 to September 2026, and Figure 3.1 shows the schedule. The first phase, in September and October 2025, built and measured the earlier version of the pipeline. It established that the speech side was the binding problem and that note quality had never been evaluated against anything. The second, from December 2025 to February 2026, checked every claim of that phase against code, retired the claims with no computation behind them and introduced the rule that a number needs a command.

The third phase produced the data, which is the transcription standard, the recordings and the hand-checked transcripts. It started in November 2025 and did not finish until September 2026, so it ran underneath every other phase and took more human hours than any of them.

The fourth phase ran the experiments on two tracks. Speech work began in February 2026 on the 12 GB machine and vision-language work in June on the 32 GB machine, so the two overlapped through the middle of the year and the second track did not

wait for the first to finish. The fifth phase, writing, began in August 2026 and produced the figures, tables and this report.

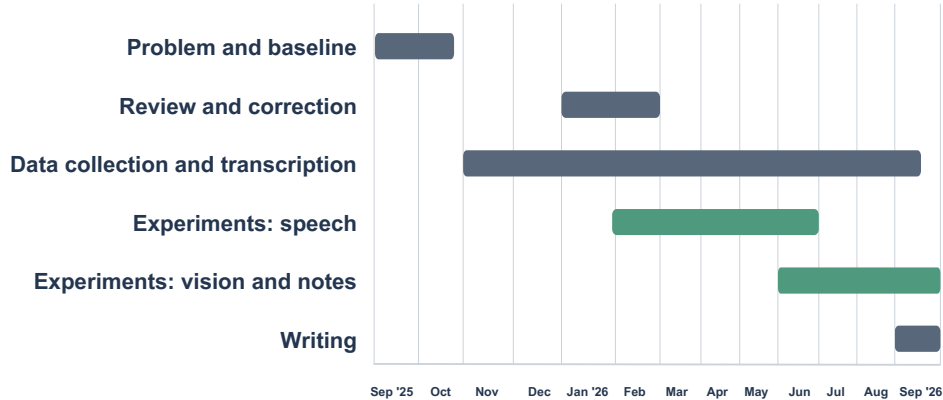


Figure 3.1: Project schedule.

Three students worked on the project under a supervisor and a co-supervisor. The resources were the two desktops described in Section 3.1.3, both already owned, the open model weights and the team’s own time; no budget was spent on cloud computing or paid annotation. Two practices mattered more than any tool. Long jobs were written as scripts that run unattended, commit their own results and push them, so a fourteen hour training run cost no attention. All planning lived in two files in the repository, one holding the ordered task list and one holding every result with the command that produces it, so work could move between machines without anything being carried in a person’s memory.

The fixed points were the draft submission and the slide submission, and work was scheduled backwards from them. When the corpus was closed at 5.15 hours the planned six hour and ten hour training runs were cancelled instead of delayed, because a result that arrives after the deadline is worth nothing.

3.7 Risk Management

Table 3.4 lists the risks identified for the project, the mitigation planned and the outcome. Several of these risks occurred, so the last column is a record rather than a prediction.

Table 3.4: Risk register, with outcomes.

Risk	Severity	Mitigation planned	What happened
Not enough transcribed speech	High	Parameter efficient fine-tuning; report both an unseen lecturer and an unseen lecture design	Occurred. The corpus closed at 5.15 h instead of ten. Both designs are reported.
Speaker leakage between training and test	High	Sort recordings by lecturer and confirm with speaker verification	Occurred. One recording was labelled as the wrong lecturer, so an early result was optimistic. Found by voice verification, recomputed, old number retired.
Training instability	High	Two random seeds for every reported configuration	Occurred. At the tuned learning rate one seed collapsed. Both seeds are reported with a stable configuration beside them.
Graphics memory exhausted	Medium	Gradient checkpointing; second machine for the large models	Occurred and was handled.
Model weights too large for the disk	Medium	Check free space before any download	Occurred. The 32 billion parameter vision-language model did not fit and was dropped.
Results that cannot be reproduced	High	A command for every number; splits and seeds written to disk	Controlled.
False statements in the generated notes	High	Ground the notes in board text, fence non-lecture material, check quotations	Partly controlled. The controls work and are measured, but a factual error in the prose was still found by reading and is reported.
Loss of work or of a machine	Medium	Version control; jobs commit and push their own results	Did not occur.
Deadline slip	High	Two machines in parallel, unattended overnight jobs, cancel rather than delay	Controlled.

3.8 Economic Analysis

The project spent no money on computing. Both desktops were already available and every model used is published under an open licence, so the real cost is human time, and most of it is transcription. Transcribing 5.15 hours of lecture speech by hand took about 31 person-hours at roughly one hour of work per ten minutes of video. Writing and verifying the 45 board records, 436 items, took about one working day, and checking 98 rebuilt boards for missing writing took 30 to 40 minutes. The graphics card time for every reported experiment was about 24 hours, which is about 8 kWh and a negligible cash amount, and storage came to about 106 GB on existing disks.

Two alternatives are worth pricing. Renting the same compute is cheap: 24 hours on a rented mid-range graphics card at typical rates of \$1 to \$2 an hour is about thirty to fifty dollars, so compute is not what makes a project of this kind expensive. Commercial transcription is the opposite. Bengali speech-to-text services exist, but they output the Bengali script, so for this purpose they would not remove any of the human work and would add a fee on top.

The benefit to be weighed is the reduction in human effort to produce a usable transcript. Producing one by hand from nothing took about six hours of work per hour of video, because off-the-shelf output was useless as a starting point. The fine-tuned model reaches a median character error rate of 15.8 per cent, which is a draft a person can correct instead of retyping. If correcting such a draft takes a third of the time, the cost falls from about six hours per hour of video to about two, which over the 8.33 hours recorded here is the difference between about fifty hours of work and about seventeen. This projection has not been tested and is not claimed as a result; measuring it properly means timing people as they correct real drafts, which is listed with the reader study in Chapter 6. What can be claimed is the input to the calculation, which is the measured error rate and the measured cost of the manual route.

A department wanting to run this on its own lectures would need one desktop with a graphics card of 12 GB or more, which is ordinary teaching laboratory hardware, and no recurring payment. Processing one lecture takes a few minutes of card time. The recurring cost is storage for the recordings and somebody to check the notes before they reach students, and that check is the part the system does not remove.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

4.1.1 Design objectives and the shape of the solution

The design follows from the objectives in Section 1.4 and the constraints in Section 3.1.3. A lecture video is the input and a lecture note containing the lecturer's words and the lecturer's board is the output. The system must run on one desktop graphics card, it must be measurable stage by stage, and it must not invent content.

One option was a single multimodal model that reads the video and writes the note. It was rejected for three reasons. A single model gives one number at the end, so when that number is poor there is nothing to inspect, whereas a pipeline gives a separate measurement for the speech, the board reconstruction, the board reading and the note. Chapter 5 shows this paying off, because the four stages turned out to contribute very differently. Keeping the stages apart also means only one model is in memory at a time, which matters on the hardware available. Finally, when the board is read into text as a separate step the note-writing model receives the board as text instead of having to remember it, so a fact in the note can be traced to the board. When one model both looks at the image and writes the note, a fact cannot be traced to its source.

Four rules were fixed before building and were not relaxed. Each stage is a separate program with a file interface, and a stage is skipped when its output already exists, so an interrupted run continues and a single stage can be replaced without touching the others. Nothing in the image is invented. The board reconstruction may leave a visible gap but may not fill one with plausible pixels. The board records used for scoring sit in a folder that the note generator refuses to read, enforced in code rather than by discipline. And every reported number is produced by a command recorded beside it.

4.1.2 The four stages

Figure 4.1 shows the whole system, with the file each stage hands to the next named on the arrow. The first two stages work on the two halves of the recording

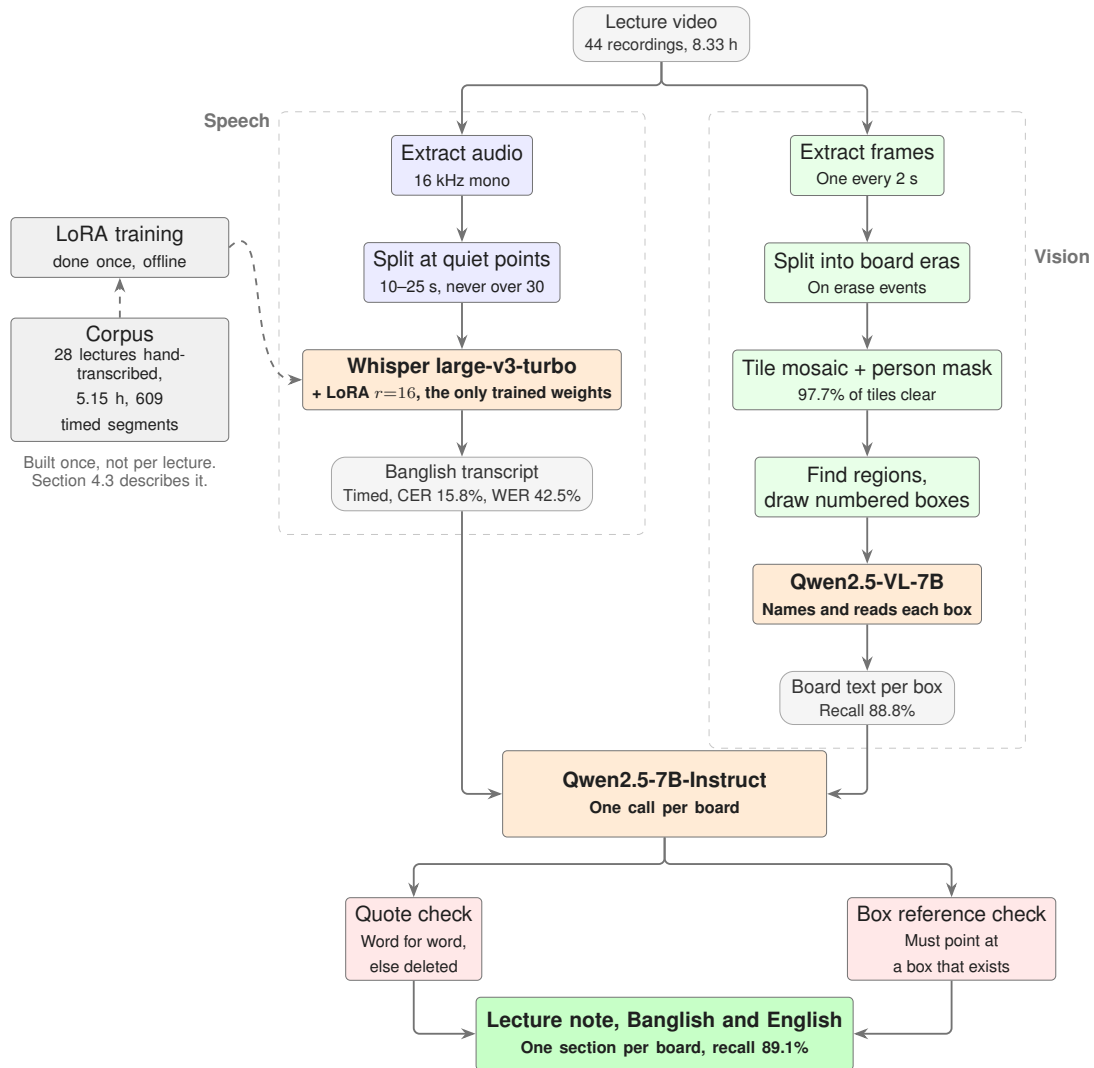


Figure 4.1: The InsightLens pipeline in full. One lecture video enters at the top and two lecture notes leave at the bottom. The speech and vision halves run independently and meet only at the note-writing model, which receives the board as text rather than as an image. Each box is a separate program that writes a file the next one reads, so any stage can be rerun on its own. The three shaded models are the only learned components, and of these only the LoRA adapter on Whisper was trained here. The grey path on the left is that training. It is drawn apart and dashed because it happens once, before any lecture is processed, and a new lecture does not need a hand-written transcript to pass through the system; Section 4.3 describes how the corpus was built. The two boxes before the output are checks that can delete text the model produced. The error rates and counts shown are the measured values reported in Chapter 5.

independently. In stage one the audio is extracted and cut into clips, a LoRA adapter trained on the corpus is attached to Whisper large-v3-turbo, and the adapted model writes a timestamped transcript in romanised Banglish. In stage two frames are extracted at a fixed interval, erase events split the lecture into eras with one board per era, and within an era each tile of the frame is taken from a moment when the lecturer was not standing in front of that tile.

The last two stages turn the board into text and then combine the two halves. In stage three the blocks of writing are found on the clean board, a numbered coloured box is drawn around each, and Qwen2.5-VL names and transcribes each numbered box. In stage four a language model receives each board’s boxes, the board text and the part of the transcript belonging to that board, and returns one section of notes that points at the boxes. Checkers then remove unverifiable quotations and bad box references, and the sections are assembled into a page.

4.2 Preliminary Design and Alternatives Considered

Several designs were built and measured before the final one was chosen. Three of them failed, and the failures are part of the argument for the design that was kept. Table 4.1 summarises the comparison, and the paragraphs below give the reasoning where the table cannot.

For the speech stage the two rejected-by-measurement options are the informative ones. Running an English recogniser and a Bengali recogniser and merging their outputs was built and measured on nine lectures. The merged output scored 0.7 percentage points above the baseline on the term measure, which a paired permutation test puts at $p = 0.32$, while its word error rate was far worse at 146.8 per cent against 82.6. Biasing the decoder towards words seen on the board, implemented as a logits processor with a strength parameter, halved term recall at every non-zero setting and truncated the transcript. Using a Bengali model and transliterating afterwards was rejected on the design and not by experiment, because it loses the English technical terms twice, first when the Bengali model spells them phonetically and again when they are romanised back.

For the board, the option that had to be abandoned was the one the first version of the project used, which removed the lecturer and filled the hole by inpainting. Inspection against the source frame found the lecturer still visible as a grey silhouette, because the brightness threshold missed the light squares of his shirt, and text he was writing at that moment lost, because the filled region was rebuilt from a single donor frame. The filled figure also showed content not yet written at that time, so it was a plausible picture of a board that never existed. Generative inpainting with a diffusion model was not attempted, because on a whiteboard a plausible fill is invented writing and a reader cannot distinguish it from the real thing. The tiled mosaic over erase separated eras was selected because every output pixel is unmodified camera output, so the only question is whether each tile found a clear moment, and that is counted exactly.

For reading the board and for writing the note, the choices were settled by the

measurements reported in Chapter 5 and are summarised in the table. One option remains untested. The code supports asking the vision-language model to draw the boxes itself, but our own box finder was kept because it is deterministic and its output can be drawn onto the real image and checked by eye.

Table 4.1: The alternatives considered at each stage. Options marked as measured were built and evaluated before being rejected.

Stage	Option	Status	Reason
Speech	Off-the-shelf Whisper	measured	CER 67.7 per cent, and it translates
Speech	Bengali model plus transliteration	rejected by design	loses English terms twice
Speech	Dual recogniser fusion	measured	+0.7 pp, $p = 0.32$, and WER far worse
Speech	Visual bias in the decoder	measured	halves term recall at any strength
Speech	Whisper plus LoRA	selected	CER 15.8 per cent on unseen lectures
Board	Best single frame	fallback	the lecturer is never absent
Board	Mask and inpaint	measured	left a silhouette and showed writing from the wrong time
Board	Generative inpainting	not attempted	would invent writing
Board	Tiled mosaic over eras	selected	every pixel is real, coverage is counted
Reading	Classical OCR	rejected	built for print, not handwriting and diagrams
Reading	Keyword prompt	measured	31.2 per cent recall, no numbers at all
Reading	Full transcription prompt	selected	88.8 per cent recall
Reading	Model draws its own boxes	untested	kept in the code, not evaluated
Notes	Free summarisation	measured	37.2 per cent, recites a textbook
Notes	Grounded prompt, keywords	measured	25.8 per cent
Notes	Grounded prompt, board text	measured	88.0 per cent, reads like a paste
Notes	Annotated note with boxes	selected	89.1 per cent and readable

4.3 Data Collection

4.3.1 Recording and transcription

Forty-four lectures were recorded in classrooms of Brac University and in other classrooms available to us, with a fixed camera pointed at the lecturer and the

whiteboard. The three people who appear as lecturers are the authors of this research, and no lecture uses slides. The recordings total 8.33 hours, the shortest 3.4 minutes and the longest 36.4 minutes.

The lecturer of each recording was confirmed by voice, not by memory. A WavLM speaker embedding was computed for every lecture and compared with reference embeddings, giving cosine similarity between 0.979 and 0.998 within a lecturer and between 0.590 and 0.899 across lecturers. This check exists because an early version of the split attributed one lecture to the wrong lecturer, which made a result look better than it was; the error is described in Section 5.1.4.

Before transcription began, a written standard was produced, and every transcript is checked against it by a validation script. Each line is a timed segment written as [M:SS-M:SS] followed by the text. Segments run 10 to 25 seconds and never longer than 30, because Whisper processes a 30 second window, and there are no gaps and no overlaps between them. Fillers such as *aaa*, *mane* and *tahole* are kept, because the goal is a transcript of the speech and not a tidied version of it. The text is ASCII only, and a table of canonical spellings fixes the most common words so that three transcribers do not write the same word three ways. Appendix A gives the standard in full. Transcription was done by hand, with every word checked against the audio by a member of the team, at a cost of about one hour of work for ten minutes of video.

4.3.2 Data Cleaning

Four cleaning problems occurred and each is handled by a script and not by hand editing. The first concerns timestamps. The segment parser accepts a timestamp only in the exact form [1:34-1:53]. In an earlier version of the corpus, 9 of 156 timestamps carried stray spaces, written for example as [4:22- 4:45]. Those lines did not register as segment boundaries, so their text joined the previous segment and ten evaluation clips carried more words than their audio. The validator now reports every unreadable timestamp and can repair them, keeping a backup of the original file. The effect of the defect on the affected result was measured rather than assumed and is reported in Section 5.1.4.

The second is incomplete files. A transcript still being written is marked by its file name and skipped until the marker is removed. Files were also delivered at one point with a placeholder extension that made them invisible to the pipeline, which was found by counting how many transcripts the loader actually saw instead of how many files existed.

The third is missing lecturer labels. Where the header was absent it was filled from the voice check rather than guessed, and any recording where the voice check was not confident was left for a person to decide. A preflight script then refuses to start training if a transcript has no matching video, if two different videos share a name, if a transcript is longer than its video, or if a lecturer label disagrees with the voice check. It runs as a hard gate before every training run, because all of these faults corrupt training silently instead of crashing.

4.3.3 Data Transformation and Integration

Each video is converted to 16 kHz mono PCM WAV, the input format Whisper expects. The human transcript gives the segment boundaries, so one clip is cut per segment, and segments longer than the model’s 30 second window are subdivided at sentence ends. Each clip becomes one training example of an audio array and its reference text. The audio is turned into a log-Mel spectrogram by the model’s own feature extractor and the reference text is tokenised by the model’s own tokeniser, so no custom preprocessing can differ between training and evaluation. For the vision side, frames are extracted at a fixed interval and scaled so that the longer side is at most 1920 pixels. A 10 second interval suffices for lecturers who move about, while a 2 second interval gives each tile five times as many chances to see the board clear for a lecturer who stands still.

The corpus was renumbered part way through the project so that lectures are grouped by lecturer. Because every result produced before the renumbering used the old names, a lookup table records the correspondence, and every script that finds a lecture’s audio or board uses the table instead of assuming the number in a file name means the same thing everywhere. The ground truth behind the earlier results is frozen in a separate folder, so those results can still be reproduced. Three sources have to line up for one lecture, which are the video, the transcript and the rebuilt boards; they are joined by lecture number through that table, and the preflight script checks the join before any training.

No dimensionality reduction is applied to the audio or the images, since the models consume raw signals. Reduction happens in three other places. Lectures without a transcript take no part in training or in the speech evaluation, which leaves 28 of the 44 for the speech experiments, while all 44 are available to the vision side. For scoring the notes only the 13 lectures with hand-verified board records can be used; the other 30 receive the same notes as demonstrations and are not scored, and the report says which is which. Within the board stage an era shorter than four frames is ignored, because too few frames means too few chances for a tile to see the board clear.

4.3.4 The splits

Three separate splits are used and they must not be confused. The validation split chooses settings. Three lectures, one per lecturer, were drawn with a fixed seed before any tuning began and recorded in a file that the data preparation script reads. They are BanglaASR2 from lecturer A, BanglaASR12 from B and BanglaASR14 from C, 34.2 minutes in total, used only to score tuning runs and locked out of every test set afterwards.

The test split gives the final result. Whole lectures are held out at random, about 20 per cent of each lecturer’s minutes, with the seed fixed at zero and the rule written down before the draw. A lecture is never cut between training and test and every lecturer appears on both sides. The draw selected BanglaASR8, 9, 11, 15, 19 and 27, which is 1.05 hours and 177 clips, leaving 4.10 hours and 22 lectures for training. Because the test lecturers are also heard in training, this design answers the question of new lectures from known lecturers, and the report labels it that way whenever it

is quoted.

The leave-one-speaker-out splits answer the harder question. Each lecturer in turn is held out entirely and the model is trained on the other two, which gives worse numbers, and both designs are reported.

One further rule protects the note evaluation. A note scored on a lecture’s board records must be written from a transcript produced by an adapter that never heard that lecture’s lecturer, so the transcripts for the scored lectures come from the leave-one-speaker-out adapters, and the builder records in each output file which adapter produced its input. Table 4.2 summarises the corpus after these steps.

4.3.5 Summary of Preprocessed Data

Table 4.2: The corpus after preprocessing.

Quantity	Value
Lecture recordings	44, 8.33 hours
Lectures with a hand-checked transcript	28, 5.15 hours of timed speech
Lectures used for vision only	16, 3.10 hours
Transcribed segments	609, median 24 seconds
Words transcribed	about 39,100
Lecturer A	9 lectures, 1.51 h, 141 segments
Lecturer B	4 lectures, 0.97 h, 39 segments
Lecturer C	15 lectures, 2.66 h, 429 segments
Training set for the final model	22 lectures, 4.10 hours
Test set for the final model	6 lectures, 1.05 hours, 177 clips
Validation lectures, never tested on	3 lectures, 34.2 minutes
Rebuilt boards	145 across 40 lectures
Boards with hand-verified content records	45, holding 436 items
Boards checked by hand for missing writing	98

Figure 4.2 shows how the recordings divide between lecturers and between training, testing and vision-only use. The imbalance is real and matters for the results, since lecturer B contributed the least speech and Chapter 5 shows that lecturer B’s test lecture improves the least. Figure 4.3 shows the segment lengths against Whisper’s 30 second window.

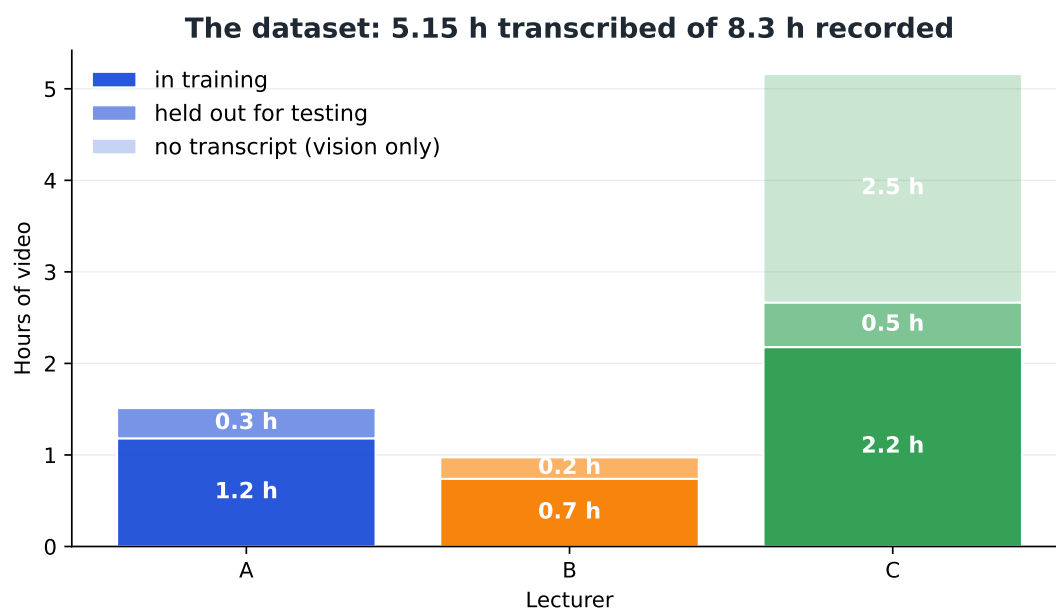


Figure 4.2: The corpus by lecturer. The darker part of each bar is training speech, the middle part is held out for testing, and the lightest part is video with no transcript.

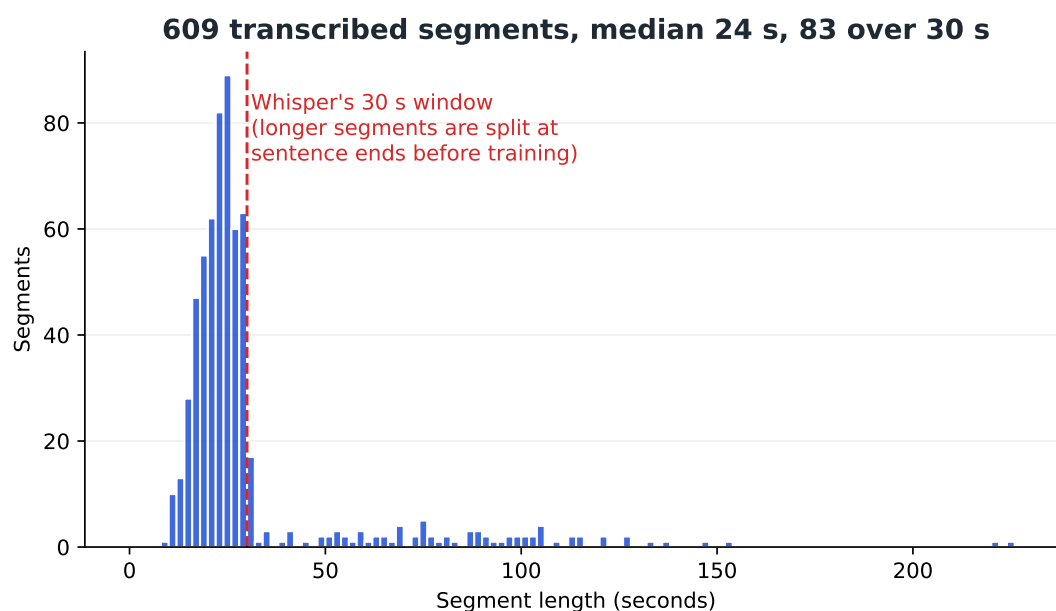


Figure 4.3: Lengths of the 609 transcribed segments against Whisper's 30 second window. Segments longer than the window are subdivided at sentence ends before training.

4.4 Implementation of Selected Design

4.4.1 Adapting the speech model

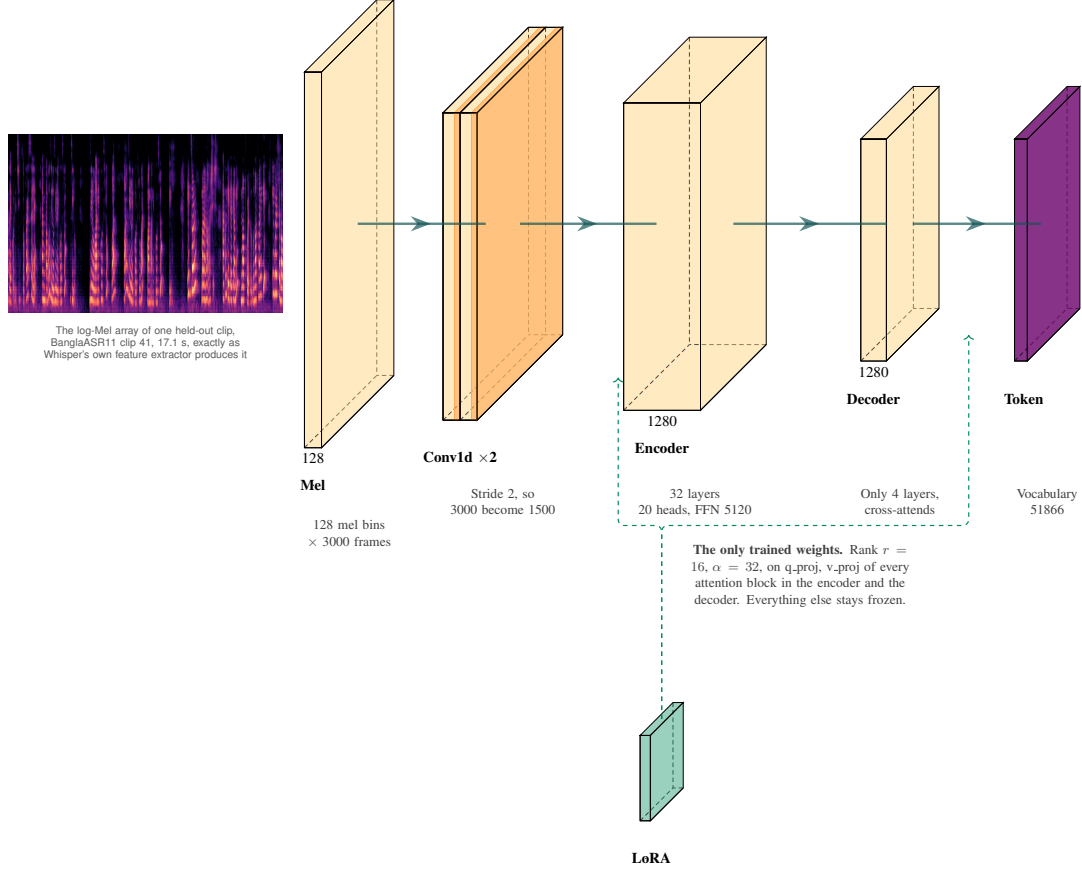


Figure 4.4: Whisper large-v3-turbo with the LoRA adapter, and one real clip passing through it. The picture on the left is the log-Mel array of clip 41 of lecture BanglaASR11, made by Whisper’s own feature extractor, so it is the array the encoder is actually given. The clip runs from 12:34 to 12:51, inside the board era shown in Figures 4.6 to 4.10, and in it the lecturer reads out the exclusive-or column of the truth table that box 2 of that board holds. Its character error rate is 24.5 per cent against the run’s median of 15.8, so this is a slightly worse than typical clip rather than a flattering one. What the fine-tuned model returned was “*so first ele ami pabo 0 0 equals to 0, 0 1 equals to 1, 1 0 equals to 1, 1 1 equals to 0. x or banana amar shesh. ekhon eitake ami not kore felbo. not mane ki? inverse ta baupar felbo.*” Every size shown is read from the model’s configuration file. The turbo model is unusual in having 32 encoder layers but only 4 decoder layers, which is what makes it fast. Only the small LoRA matrices were trained; the 809 million parameters of the base model were never updated.

Figure 4.4 shows the model and one real clip passing through it. The base model is whisper-large-v3-turbo, an encoder-decoder transformer with 809 million parameters [1], and its weights are frozen. LoRA adapters are attached to the query and value projection matrices of the attention layers, with rank 16 and scaling factor 32 [2]. Training uses a batch size of 8, 8 epochs, bfloat16 precision and gradient

checkpointing on the 12 GB card, which reduces the memory needed from about 18 GB to 6.1 GB at the cost of recomputing activations. Every run is done with two random seeds and both are reported.

The reported numbers use plain greedy decoding. The loop safeguard from the Whisper paper is also available and is reported beside the main result. A hypothesis whose gzip compression ratio exceeds 2.4 is decoded again with sampling at rising temperatures until the output stops looping. It reads only the hypothesis, never the reference, and is applied identically to the base and the adapted model.

The settings come from a plan committed to version control before any tuning run, which fixes the validation lectures, the order of the search and the rule for accepting a winner. The plan has five stages, run in order, each starting from the winner of the previous one.

1. Learning rate, over 5e-4, 1e-3 and 2e-3.
2. LoRA rank, over 8, 16 and 32, with the scaling factor at twice the rank.
3. The adapted layers, comparing the query and value projections against all four attention projections.
4. The number of epochs, taken from the validation loss curve.
5. Seed stability, by repeating the winner with a second seed.

Everything else is held fixed. The score is the median per-clip character error rate on the validation clips, and a candidate replaces the default only if it beats the default by more than the spread between two seeds of the same configuration, which is the plan's way of refusing to chase noise. The procedure was run twice, first as a rehearsal on 2.69 hours of transcript to check that the machinery worked, then on the full corpus once transcription closed; the second run supersedes the first and Section 5.1.2 reports it.

4.4.2 Rebuilding the board

The steps are shown in Figure 4.5. The amount of ink on the board is tracked across frames, and when a large fraction of it disappears between consecutive frames the board has been wiped, so the frames between two wipes form one era. Within an era the frame is cut into small overlapping tiles. A tile counts as clear in a given frame when almost none of it is covered by the lecturer, and for each tile independently the pipeline takes the latest frame in which that tile was clear. The chosen tiles are blended back together with a raised-cosine weight so that the seams do not show.

The method works because the lecturer occupies one place at a time, which is also its stated limit. It needs the lecturer to move. Where a tile never finds a clear moment the residue is left visible and reported instead of filled in.

Locating the lecturer has three implementations and the third is used. The simplest takes whatever differs from the usual appearance of the board, which fails when the lecturer stands still, because then he is part of the usual appearance, and fails

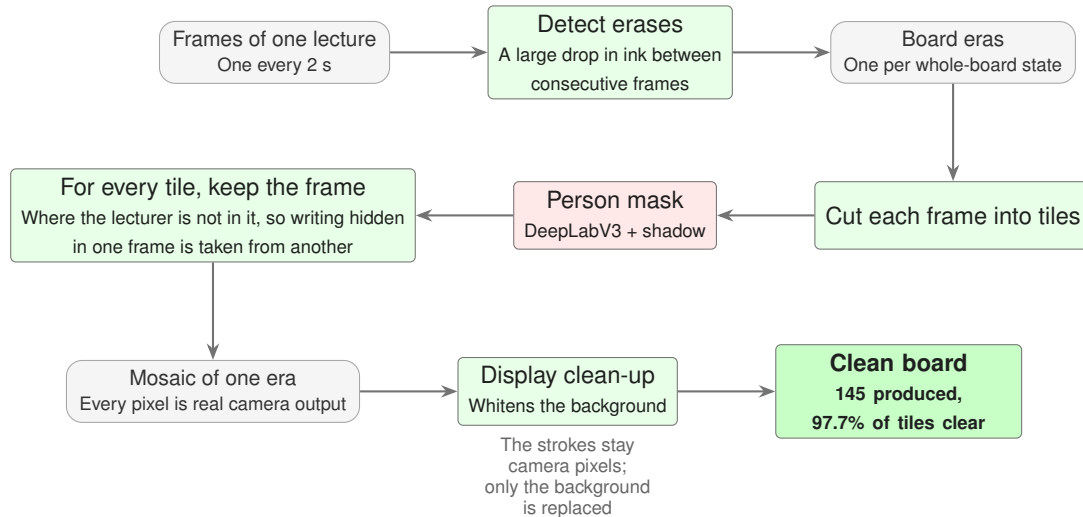


Figure 4.5: How one clean board is built from a lecture video. Nothing in this chain generates pixels: every part of the finished board was photographed by the camera at some moment, and the only step that changes a pixel is the display clean-up at the end, which replaces the background but leaves the strokes alone.

again when the camera’s automatic exposure darkens a frame, because then every pixel differs. The second uses a pretrained DeepLabV3 network with a ResNet-101 backbone and takes its person class frame by frame [54]. The third adds a shadow mask, computed as board darker than usual after dividing out each frame’s exposure gain, with pen-stroke-thin marks eroded away so that writing is not mistaken for shadow. The third recovers writing the first lost, including a diagram drawn at the end of an era and seven of the eight digits of a binary number that the old mask reduced to four.

A separate clean-up step flattens the background and fades everything not clearly darker than its surroundings to white, in the manner of the whiteboard mode of a document scanner, and blanks the pixels the mosaic flagged as the lecturer. Because it replaces background pixels, a cleaned board is not unmodified camera output. The strokes are camera pixels and the background is not, and the report says so wherever a cleaned board is shown. Section 5.2.2 reports the measured effect of this step on how well the vision-language model reads the board, which was not the expected one.

Figure 4.6 shows why one frame is not enough, and Figures 4.7 and 4.8 show the board this era produces, before and after clean-up.

4.4.3 Numbered boxes and the vision-language model

Figure 4.9 shows this stage on a real board. Our own code, not the model, finds the blocks of writing. It groups dark pixels, strips the board frame, drops specks and rejoins boxes that are parts of the same written line, producing at most ten boxes per board. Each box receives a number and a colour from a fixed list of ten, and the notes later refer to a box by both, for example the purple box 5. Figure 4.10 is

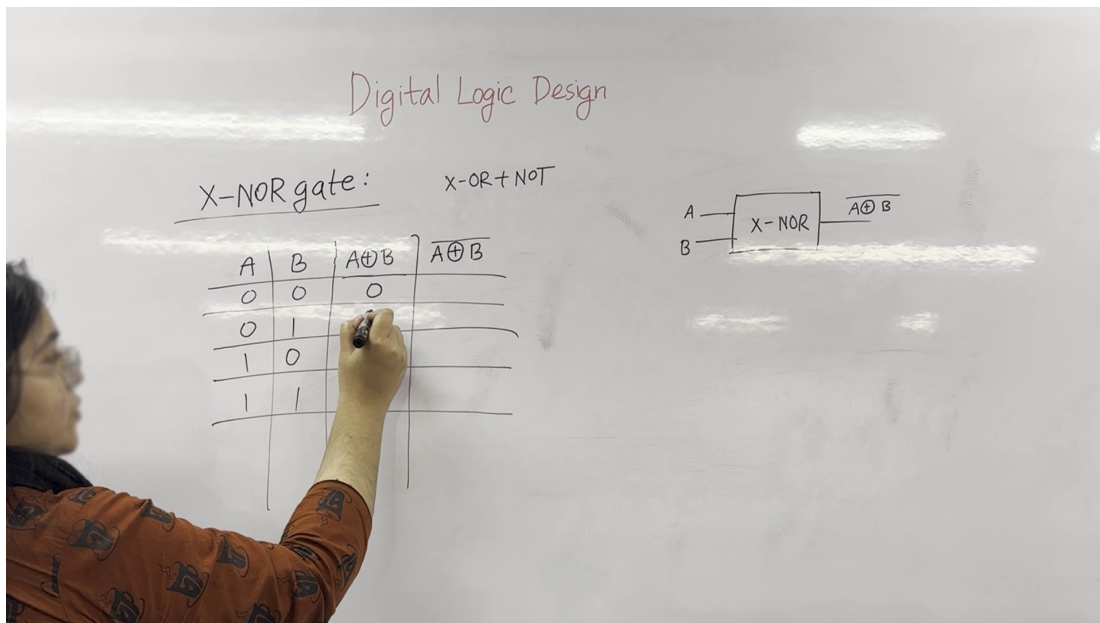


Figure 4.6: One frame of the digital logic lecture, BanglaASR11, at 12:40. The lecturer is filling in the exclusive-or column of the truth table, and her arm and body cover the left half of the board while she does it. The last two columns do not exist yet, so no single frame of this era shows the finished board. Figure 4.7 is what the reconstruction recovers from the era as a whole.

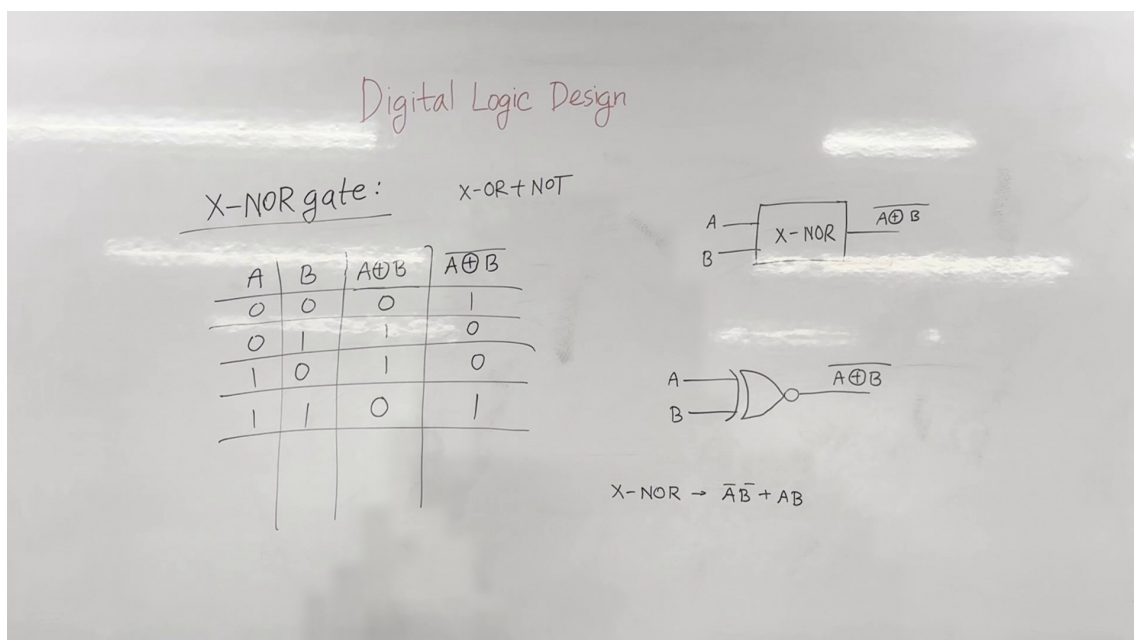


Figure 4.7: A rebuilt board from lecture BanglaASR11. Every pixel comes from a real frame of the video. The horizontal bands are glare from the ceiling lights, which is present in every frame and therefore cannot be removed by choosing a different one.

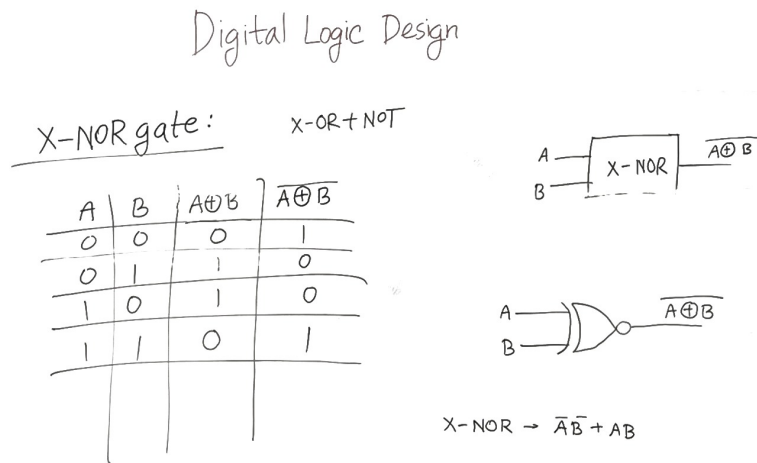


Figure 4.8: The same board after the display clean-up. The background has been replaced, so this image is enhanced and not unmodified camera output, while the strokes themselves are camera pixels.

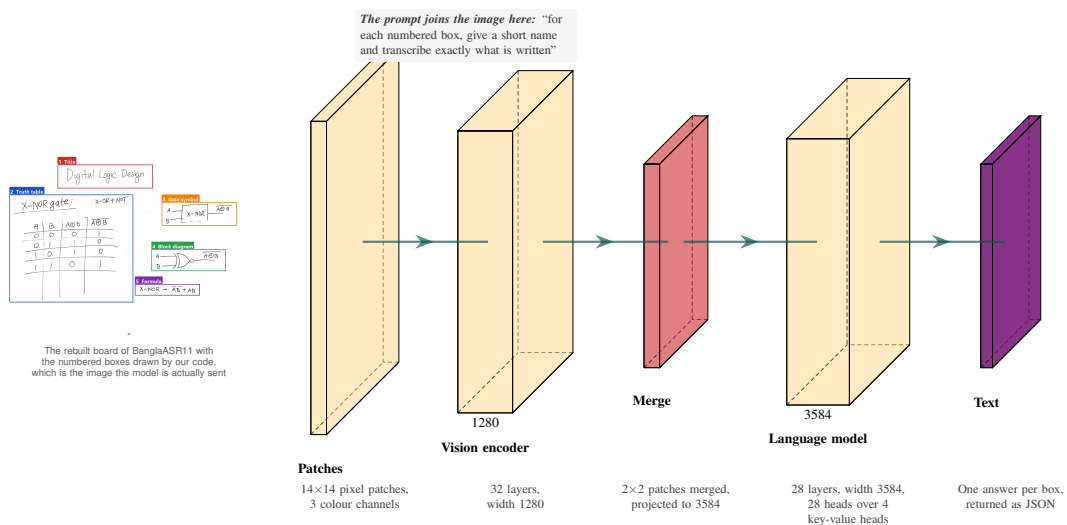


Figure 4.9: Qwen2.5-VL-7B reading a board, with the real board of lecture BanglaASR11 as the input. The numbered boxes on that image were drawn by our own code, not by the model; the model is asked to name and transcribe each number in turn, which is Set-of-Mark prompting [5]. For this board it returned *box 1* "Title": *Digital Logic Design*, *box 2* "Truth table": *X-NOR gate, X-OR + NOT, A, B, $A \oplus B$, $\overline{A \oplus B}$* and *box 5* "Formula": *$X-NOR \rightarrow \overline{A} \overline{B} + AB$* . Sizes come from the model's configuration file. The vision encoder and the language model are one network, since image patches become tokens in the same 3584-dimensional space as the text, which is why the model can be asked about a specific numbered region in words.

a finished example.

The numbered boxes are drawn onto the clean board and the image is given to Qwen2.5-VL-7B-Instruct with a prompt asking, for each numbered box, for a short name and a full transcription of what is inside it. This is Set-of-Mark prompting [5] with the marks supplied by our code. Boxes the model reports as empty are dropped.

Two defects were found during the real runs and both are handled in code. On one board the model read the content correctly, then tried to draw a diagram's diagonal and emitted the same backslash line 512 times, giving 42,037 characters; the reader now collapses runs of identical lines and records how many it dropped, which matters because that text is pasted into the note prompt. Separately, boards carrying code make the model emit unescaped quotation marks that break the JSON it is asked to return, and without a fallback such a board loses all of its boxes, so a salvage parser reads the fields one at a time when the JSON cannot be parsed.

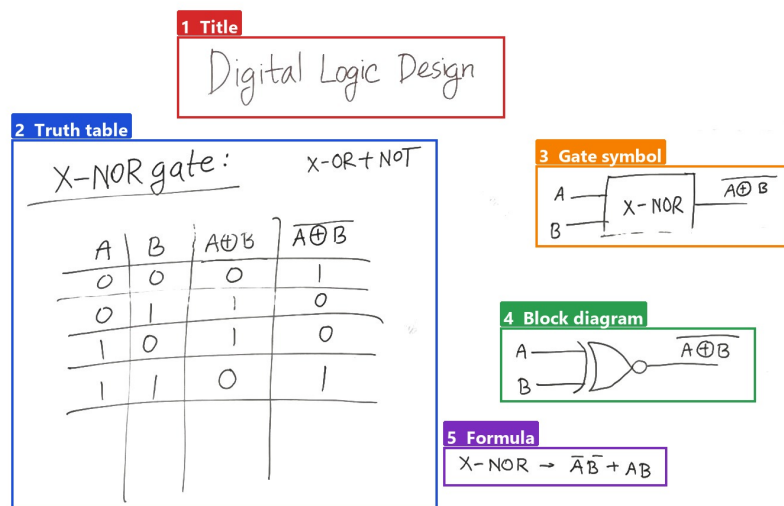


Figure 4.10: The board of Figure 4.8 with the numbered coloured boxes drawn by our code and named by the vision-language model. The generated notes refer to these numbers and colours. The naming is not perfect and the figure is left as the model produced it: box 3 is the block diagram and box 4 is the gate symbol, and the model has given each of them the other's name. Both names describe something that is on the board, which is why an error of this kind survives a recall measure that scores content rather than labels.

4.4.4 Writing the note

Figures 4.11 and 4.12 show the model and the steps around it. The note is written by Qwen2.5-7B-Instruct, one call per board. Each call receives the board's boxes with their names and transcribed text, and the part of the transcript falling between that board's first and last frame, so the model sees the words spoken while that board

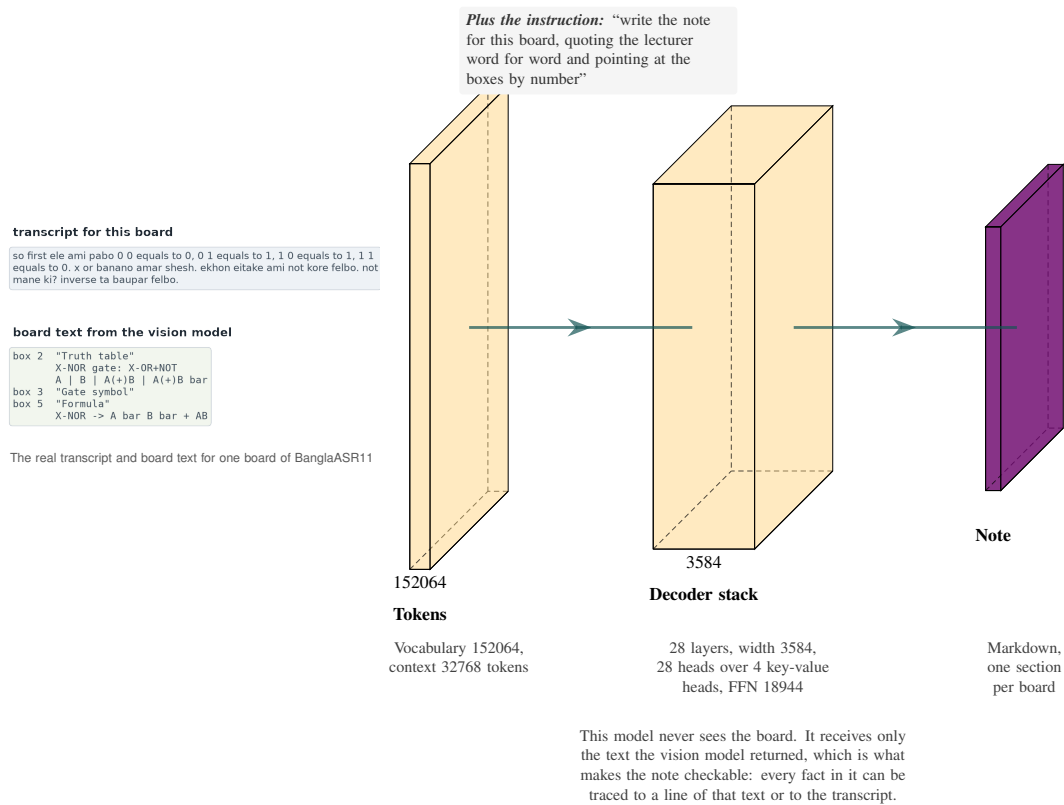


Figure 4.11: Qwen2.5-7B-Instruct writing one section of a note. The panel on the left is the real input for the board of Figure 4.10, that is the transcript of the minutes the board was on the wall and the box text the vision model returned for it. What came back includes “*Truth Table (Box 2): the truth table for the X-NOR gate is shown above. The output is 1 if both inputs are the same, both 0 or both 1, and 0 if the inputs are different.*” This model has no image branch at all. It never sees the board, only the text read off it, which is the property that makes the output checkable.

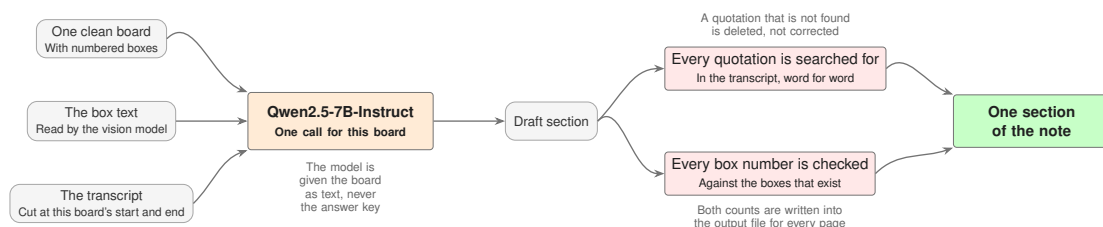


Figure 4.12: One board and the matching minutes of transcript becoming one section of a note. The two checks on the right can delete text the model produced, and the number of deletions is written into the output file for every page.

was on the wall. A final call produces the title, the takeaways and the self-check questions for the whole lecture.

The prompt asks for a section that walks the reader through the board, refers to the boxes by number and colour, and quotes the lecturer between two and five times. Material that is standard for the topic but that the lecturer did not say is allowed only inside a section headed Background, which the page builder renders as a dashed, labelled aside. This is the controlled version of the failure that produced the wrong textbook definition in the first version of the pipeline. The material is still generated, but it is fenced and labelled rather than passing as something the lecturer said.

Three checks then run over the generated text. The quote checker searches the transcript for every block quotation and deletes any that is not there word for word, writing the count of kept and deleted quotations into the output file. A second counter flags quoted spans of four or more words inside ordinary paragraphs that match neither the transcript nor the board, but only flags them, since most turn out to be code or translated lines rather than errors. The box reference check removes any reference to a box number the board does not have and counts them. The placeholder strip removes the prompt's own example text when the model copies it into the notes instead of leaving the quotation out, a failure found by reading the output.

Each version is asked for in a single language rather than the mixed style of the mock-up, and the two differ in how the lecturer is quoted. The Banglish note quotes the lecturer's real words, so the quote checker applies to it. The English note quotes the lecturer in English translation, and a translation cannot be matched against a Banglish transcript, so the checker does not run on the English files. Compliance with the language instruction is partial: the explanatory prose of the Banglish note is often written in English even when the quotations are not, so the labels name the language the section was requested in rather than a guarantee about every sentence. Every output file records which of the two it is, and Chapter 5 states the resulting claim precisely. The English note is produced by writing the section in Banglish first and then translating it with the same model, keeping the Markdown structure, the tables, the box numbers and the colours. A defect in this route was found and fixed. The translation left 97 section labels in Banglish across the thirteen scored files and returned three summary lines in capital letters, both repaired by a mapping script that does not call the model, since the labels are a fixed set written by our own prompt.

The sections are finally assembled into Markdown and into one self-contained HTML page with the board images embedded, coloured tags for the box references and click-to-reveal answers for the self-check questions.

4.4.5 Running the system, and how it is evaluated

The whole pipeline is one command that takes a video and produces the notes. It runs each stage as a separate process, skips any stage whose output already exists, and writes the time each stage took to a file beside the outputs, which is what makes the run times in Section 3.3 measurements rather than estimates.

Speech is evaluated by computing word and character error rate for each test clip separately and reporting the median, because a single runaway output can push a mean above 100 per cent. The base model and the adapted model are run on exactly the same clips, so the comparison is paired and is tested with a Wilcoxon signed-rank test and a sign test, and the number of runaway clips is reported for both. Two spelling tolerant variants are also computed, with the rules committed before any rescoring. The first maps both sides to the transcription guide’s canonical spellings and collapses repeated vowels, and the second additionally counts a word as correct at 80 per cent character similarity, so words of three letters or fewer must still match exactly.

For the board reconstruction two different things are counted and must not be confused. The fraction of tiles that found a completely clear view is computed by the reconstruction itself and measures coverage and not whether writing survived. Whether writing survived was answered by a person, who viewed each rebuilt board beside two real frames from the same time and said whether anything written was missing and what it was.

Board reading and note quality are both scored by recall of items from the hand-verified records. A record is a list of the things written on one board, each labelled as a number, name, code fragment, term or phrase. Recall is the right measure and not F1, because good notes legitimately contain a great deal that was not on the board, so precision would punish the behaviour we want. Items of three characters or fewer must match as whole words, so that *CS* is not found inside *economics*. The records were drafted from the board images and then checked by hand, one board at a time, without the checker seeing any model output, and the folder holding them is refused by the note generator in code.

Three reporting rules apply throughout Chapter 5. Medians are reported, not means. The direction of every difference is stated beside its p-value. Counting by board is reported alongside counting by item, because the item count is dominated by a few dense boards and the two can disagree.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Each stage of the pipeline is evaluated on its own, because a single end-to-end number would hide which part works. Table 5.1 collects the main results and the sections below give the conditions and the caveats for each.

Table 5.1: The main results, one per stage.

Stage	Measure	Result
Speech, six unseen lectures	Median per-clip character error rate	67.7% to 15.8% and 16.0% in two seeds
Speech, unseen lecturer	Median per-clip character error rate	72.8% to 50.3%, all six runs significant
Board reconstruction	Tiles finding a clear view, 35 boards	median 97.7%
Board reconstruction	Human check, 98 boards	82 complete, 9 with writing genuinely lost
Board reading	Recall of 349 hand-verified items	31.2% to 88.8% by changing the prompt
Generated notes	Recall of 349 hand-verified items	37.2% to 89.1%
Generated notes	Blind preference of 20 readers	preferred by 15 of the 19 who chose

5.1.1 Speech recognition: the final result

The corpus was closed at 28 transcripts and 5.15 hours. Whole lectures were held out at random, 20 per cent of each lecturer’s minutes with the seed fixed at zero, giving 22 training lectures totalling 4.10 hours and six test lectures totalling 1.05 hours and 177 clips. The three validation lectures used for tuning were locked out of the test set.

Table 5.2: Median per-clip error on the 177 test clips, off-the-shelf against fine-tuned, greedy decoding throughout. The last row is the row above it with a different random seed.

Learning rate	Seed	CER	WER	Runaway clips	Better of 177	Wilcoxon
off-the-shelf	–	67.7%	93.9%	13	–	–
1e-3	42	15.8%	42.5%	1	164	p < 1e-30
1e-3	1	16.0%	41.7%	1	167	p < 1e-30
2e-3 (the tuned choice)	1	16.2%	41.7%	2	168	p < 1e-30
2e-3 (the tuned choice)	42	118.7%	189.8%	82	35	worse

The character error rate falls from 67.7 per cent to 15.8 and 16.0 per cent and the word error rate from 93.9 per cent to 42.5 and 41.7 per cent. Two seeds agree to within 0.2 points and the fine-tuned model is better on 164 and 167 of the 177 clips, so about three quarters of the character error is removed. Figure 5.1 shows the comparison and Figure 5.2 shows it clip by clip.

The instability in the last row of Table 5.2 has to be reported with the result. At 2e-3, the learning rate the pre-registered procedure chose on the validation lectures, one of the two seeds collapsed: 82 of the 177 clips produced runaway output and the character error rate reached 118.7 per cent, which is worse than doing nothing. The data, the split and every other setting were identical to the seed that reached 16.2 per cent, so the 16.2 per cent figure is never quoted in this research without its failed twin. The same configuration has shown this fragility before, since LoRA rank 32 and all four attention projections both diverged during tuning.

The 1e-3 rows also require a disclosure. The pre-registered procedure selected 2e-3, and the runs at 1e-3 were launched after the divergence had been seen, so that choice was not blind and 1e-3 is not presented as the pre-registered winner. What can be claimed is narrower. The runner-up rate from the same tuning table is stable across two seeds on the test set while the chosen rate is not. The selection rule itself was not changed and remains recorded in the committed tuning plan.

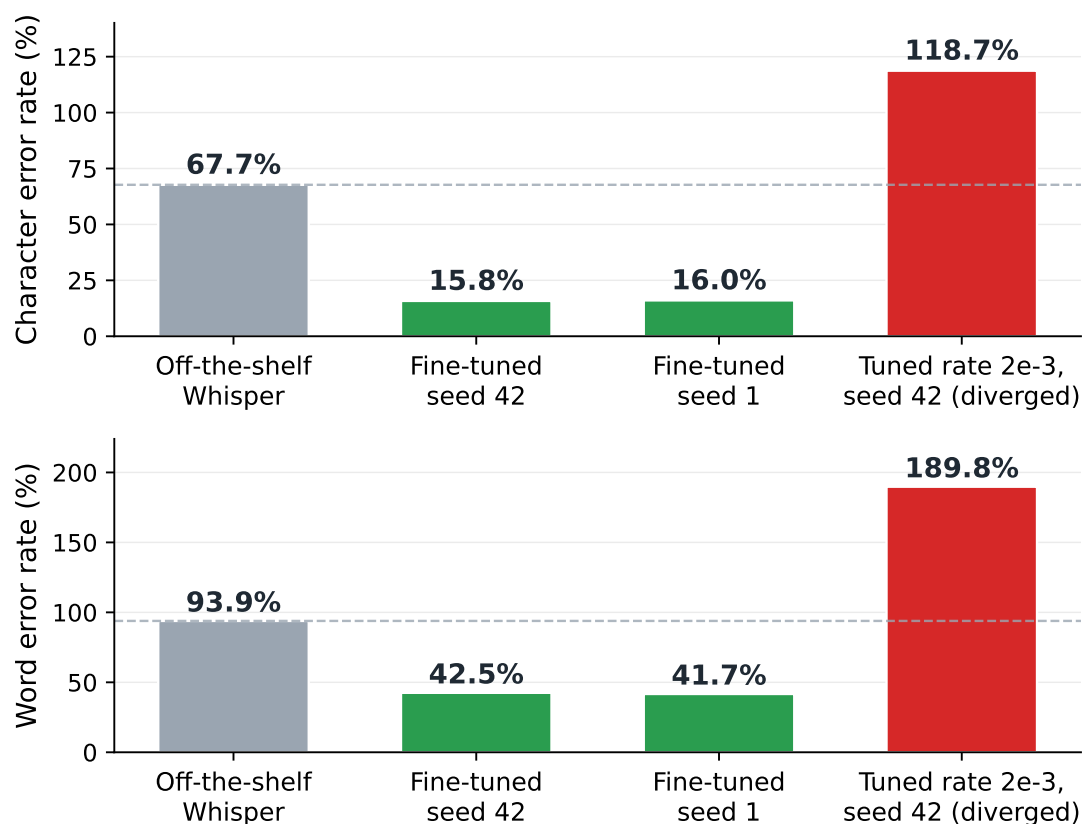
5.1.2 How the training settings were chosen

The tuning procedure was run on the full corpus, scored only on the three validation lectures and never on the test lectures. Off-the-shelf Whisper on those validation clips gives a character error rate of 73.8 per cent and a word error rate of 97.6 per cent. Table 5.3 gives every stage of the search.

The procedure chose a learning rate of 2e-3 with rank 16, the query and value projections and 8 epochs. That configuration is 8.6 character error points better than the starting point, and the acceptance rule required it to beat the default by more than the 5.4 point spread between two seeds.

Three findings follow from this table. More adapter capacity hurts at this learning rate, since rank 32 diverged outright and adapting all four attention projections

Banglish ASR: 177 clips from six lectures held out of training (4.10 h of training speech)



Lower is better. Both stable seeds use learning rate 1e-3. The red bar is the rate the pre-registered tuning chose (2e-3): its other seed reached 16.2%, this one collapsed into repetition loops on 82 of 177 clips.

Figure 5.1: The final speech result. Median per-clip error before and after fine-tuning for both stable seeds, with the diverged seed shown beside them rather than left out.

diverged as well, so five hours of speech does not support more trainable parameters. Validation loss is a poor guide to the metric that is reported. The loss is lowest at 5 epochs, at 1.666 against 1.801 at 8 epochs, so stage 4 retrained there and the character error rate got worse, from 33.6 to 39.4 per cent. And the margin is not large next to seed noise, at 8.6 points ahead of the default with a 5.4 point spread from a single seed change, measured on 34 minutes of validation speech. It passes the rule written in advance while remaining a modest effect, which is consistent with the divergence that the same configuration later showed on the test set.

A rehearsal of the procedure had been run earlier on 2.69 hours of transcript to check that the machinery worked. It chose the same learning rate and produced the same two divergences.

An earlier result in this project, on a smaller model, only beat its baseline when

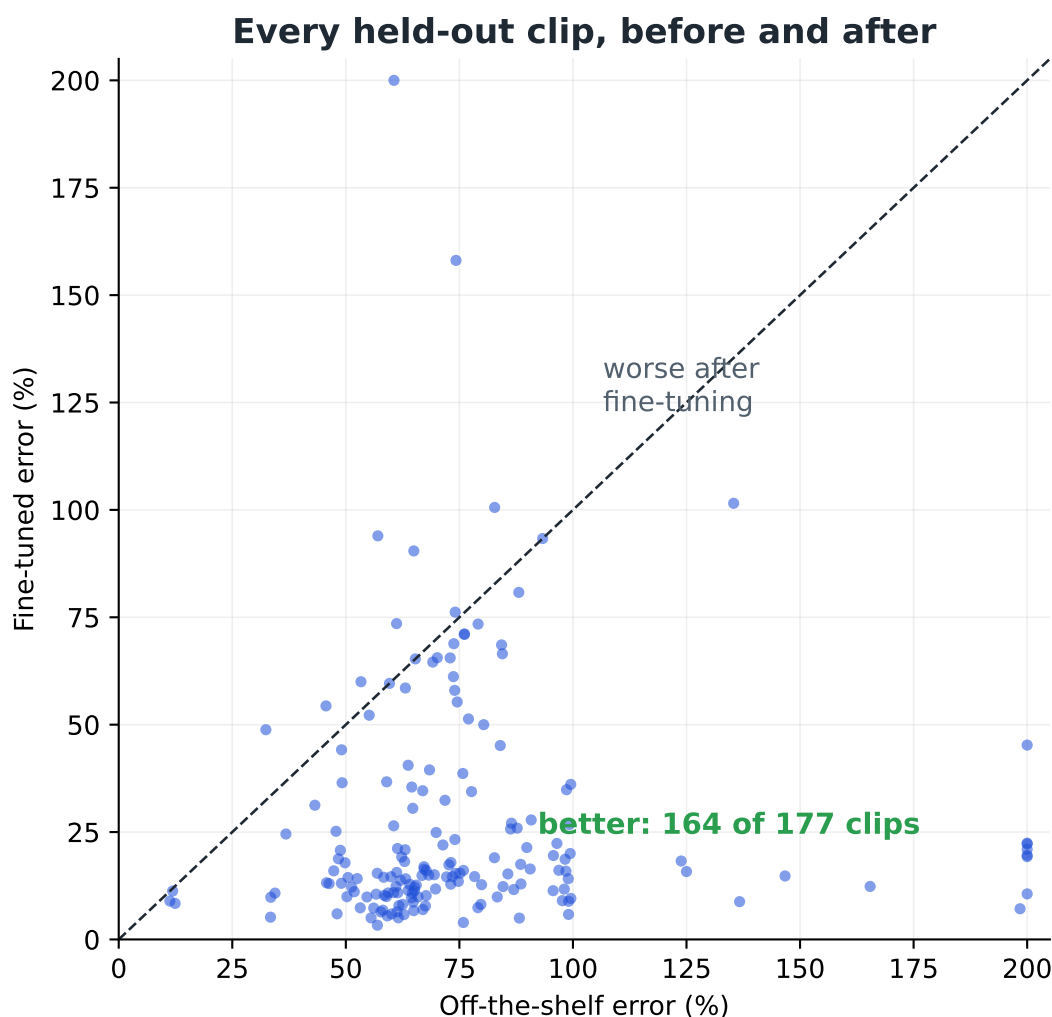


Figure 5.2: Each point is one test clip, with its error before fine-tuning on one axis and after on the other. Points below the diagonal improved. The points at the extreme right are clips where the off-the-shelf model fell into a repetition loop.

Whisper’s compression-ratio loop safeguard was switched on, so the obvious challenge to the present result is that the safeguard is doing the work. Rescoring both final models with the safeguard gives 15.8 per cent for seed 42 and 16.0 per cent for seed 1, identical to the greedy figures to the first decimal, because the fine-tuned model loops on one clip in 177 and the safeguard has nothing to rescue. Off-the-shelf Whisper loops on thirteen. All figures reported in this research are the plain greedy ones.

5.1.3 Where the speech gain is uneven, and whether it is only spelling

The improvement is not evenly spread, and the pattern follows the training data. Table 5.4 gives the error rate for each test lecture, and Figure 5.3 shows the same numbers as a chart.

Table 5.3: Pre-registered hyperparameter tuning on the full corpus, scored on the three validation lectures. Each stage starts from the winner of the previous one. The rows marked diverged produced output worse than the off-the-shelf model.

Stage	Run	lr	rank	layers	epochs	val CER	val WER
1 learning rate	5e-4	5e-4	16	q,v	8	37.9%	60.0%
1 learning rate	1e-3 (default)	1e-3	16	q,v	8	42.2%	61.9%
1 learning rate	2e-3	2e-3	16	q,v	8	33.6%	54.5%
2 LoRA rank	rank 8	2e-3	8	q,v	8	40.1%	64.1%
2 LoRA rank	rank 32	2e-3	32	q,v	8	126.6%	200.0%
						di-	
						verged	
3 adapted layers	four projec-	2e-3	16	q,k,v,o	8	134.7%	174.4%
	tions					di-	
						verged	
4 epochs	5 epochs	2e-3	16	q,v	5	39.4%	59.6%
5 stability	seed 1	2e-3	16	q,v	8	39.1%	59.6%

Table 5.4: Median character error rate per test lecture, seed 42, with the amount of that lecturer’s speech in the training set.

Test lecture	Lecturer	Clips	Off-the-shelf	Fine-tuned	That lecturer in training
BanglaASR8	A	14	68%	15%	1.2 h
BanglaASR9	A	41	67%	12%	1.2 h
BanglaASR11	B	46	70%	52%	0.7 h
BanglaASR15	C	8	88%	12%	2.2 h
BanglaASR19	C	12	65%	12%	2.2 h
BanglaASR27	C	56	73%	16%	2.2 h

Every lecture improves, but lecturer B’s lecture improves by far the least, ending at 52 per cent where the others reach 12 to 16, and lecturer B contributed the least training speech at 0.7 hours. With one test lecture per lecturer at the low end this is an observation rather than a controlled result, since it cannot separate less data from this lecturer from this lecturer being harder to transcribe. It is the most likely explanation and the clearest argument for collecting more speech from under-represented lecturers.

Banglish has no standard spelling, so a reasonable challenge is that the word error rate is inflated by *aami* against *ami*. Two spelling tolerant variants were computed on the same saved outputs with the rules committed before any rescoring. These come from the six leave-one-speaker-out runs and not from the final model, because that was the headline at the time the rules were fixed. Over the mean of the six per-run medians, plain word error rate moves from 95.3 to 75.7 per cent between the base and the fine-tuned model, the spelling-fair variant from 95.0 to 74.8, and the fuzzy variant that accepts a word at 80 per cent character similarity from 92.7 to 68.2. The canonical spelling table therefore moves the word error rate by about one point and near-miss spellings by about seven more, so most of the word error is

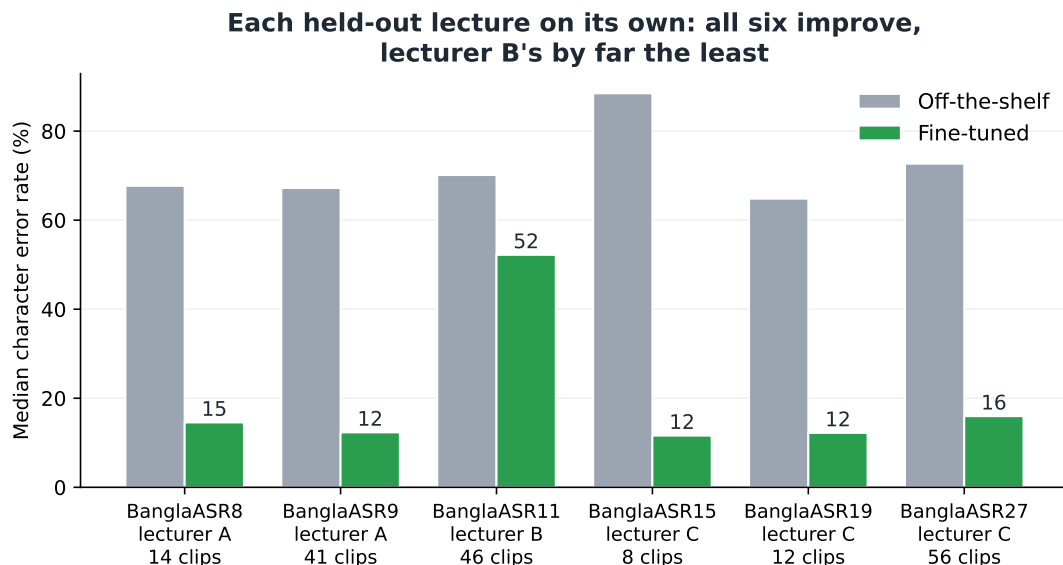


Figure 5.3: Median character error rate for each test lecture before and after fine-tuning. Every lecture improves, and lecturer B's lecture improves least.

genuine recognition error. The fine-tuned model is better in all six runs on all three measures, with p between $2.2\text{e-}16$ and $2.7\text{e-}06$.

5.1.4 An unseen lecturer, and a correction to an earlier result

The result in Section 5.1.1 holds out whole lectures from lecturers who are also heard in training. A harder question is what happens on a lecturer the model has never heard, so each lecturer was held out in turn and the model trained on the other two, with two seeds per fold and plain greedy decoding. Table 5.5 gives the three folds.

Table 5.5: Leave-one-speaker-out. Each row holds out one lecturer entirely. The two tuned figures in each cell are the two training seeds.

Held out	Trained on	WER		CER		Wilcoxon
		base	tuned	base	tuned	
A, 172 clips	B + C, 82.9 min	97.5%	76.2 / 75.8%	74.2%	51.8 / 49.9%	$p < 1\text{e-}10$
B, 184 clips	A + C, 80.0 min	93.8%	73.5 / 74.8%	70.0%	48.8 / 52.1%	$p < 1\text{e-}08$
C, 75 clips	A + B, 114.0 min	94.6%	74.5 / 79.8%	74.2%	45.2 / 54.0%	$p < 1\text{e-}05$

Averaged over the six runs the character error rate falls from 72.8 to 50.3 per cent and the word error rate from 95.3 to 75.7 per cent. All six runs are significant under plain decoding and runaway clips fall or stay level in every fold. Fine-tuning on 80 to 114 minutes of transcribed Banglish from two lecturers therefore cuts the character error on a third, unheard lecturer by about a third, for every one of the three lecturers and in both seeds. It does not follow that this is usable speech recognition, since a 50 per cent character error rate is still very high, nor that the

finding holds across many speakers, since three is a small number. Both designs are labelled every time they appear: Section 5.1.1 is new lectures from known lecturers on 4.10 hours, and this one is an unseen lecturer on 80 to 114 minutes.

An earlier version of this work reported a word error rate falling from 96.1 to 81.8 per cent on a speaker never seen in training. That claim was wrong. The original nine transcripts carried no lecturer label, so the lectures were labelled by hand and one was attributed to the wrong lecturer. That lecture was in the training set and its lecturer was the test speaker, so 47 clips of the supposedly unseen speaker had been trained on. The error was found by computing speaker embeddings, which separate the groups without ambiguity at 0.99 to 1.00 similarity within a lecturer and 0.75 to 0.80 across. The result was recomputed on the corrected split, the old number was retired, and the voice check is now a standing part of data preparation.

A second check concerns the malformed timestamps described in Section 4.3.2. Both models were scored on the same clips, so the comparison is fair, but the effect was measured rather than assumed. Without the ten affected clips the result is 72.2 per cent to 49.6 per cent character error against the published 72.8 to 50.3, so the defect made the published figure slightly pessimistic and the published figure stands.

5.1.5 Rebuilding the board

The reconstruction is measured first by coverage, the fraction of tiles that found a moment when the lecturer was not in front of them. Across the 35 boards of the first nine lectures the median board has 97.7 per cent of its tiles fully clear, the mean is 96.5 per cent and the range is 87.5 to 100. Six boards reach 100 per cent, 26 of the 35 reach 95 per cent or better, and only two fall below 90. Over all 145 boards produced in the project the median is 100 per cent and the mean 98.4, with the worst board at 76.6.

Short eras give a tile fewer chances to be seen clear, and six eras are under eight frames. A lecturer who stands in one place is the other cause, and the worst lecture in the set is the one where the lecturer holds a single position longest. That second cause was addressed. Extracting frames every 2 seconds instead of every 10 raised the number of his boards reaching 95 per cent clear from 1 to 5 out of 10 and improved the worst board from 59 to 88 per cent.

Coverage measures the method and not the result, and it should not be read as fidelity. A tile can be clear while the writing is still missing, because the writing may never have been visible in any frame. Answering that needs a person, so every one of the 98 boards of the newer lectures was checked by hand, with each rebuilt board shown beside two real frames from the same period and the question being whether the board contains everything that was written. Table 5.6 gives the counts.

The nine real losses are small. Seven are a single token or one diagram, such as a label t4, the number 100 or the word Hybrid, and one board is the bad case, where a whole subnetting tree is absent. Counting only the boards whose source frames were usable, which drops the four mid-erase eras and the three out-of-focus ones, the reconstruction is complete on 82 of 91 boards, or 90.1 per cent. The four mid-erase cases have a different cause and a different fix. The era boundary was placed while the lecturer was still wiping, so every frame in that era shows a half erased board,

Table 5.6: Human check of 98 rebuilt boards.

Verdict	Boards	
Complete	82	83.7 per cent of 98
Something missing	16	
writing genuinely lost	9	9.2 per cent
era cut while the board was being wiped	4	half erased in every frame of that era
camera out of focus for the whole era	3	unreadable in the source video

and holding the era open until the wipe finishes would correct it.

These three sets of numbers count different things and should not be added together. Coverage counts tiles, the human check counts whether writing survived, and the recall figures below count whether a model could read the writing back.

5.1.6 Reading the board and generating the notes

The same Qwen2.5-VL-7B model was run twice on the same board images, changing only the prompt, and Table 5.7 gives the result. The first asks for keywords, which is what the original pipeline did. The second asks for a full transcription of everything on the board, with every number exactly as written and [illegible] rather than a guess.

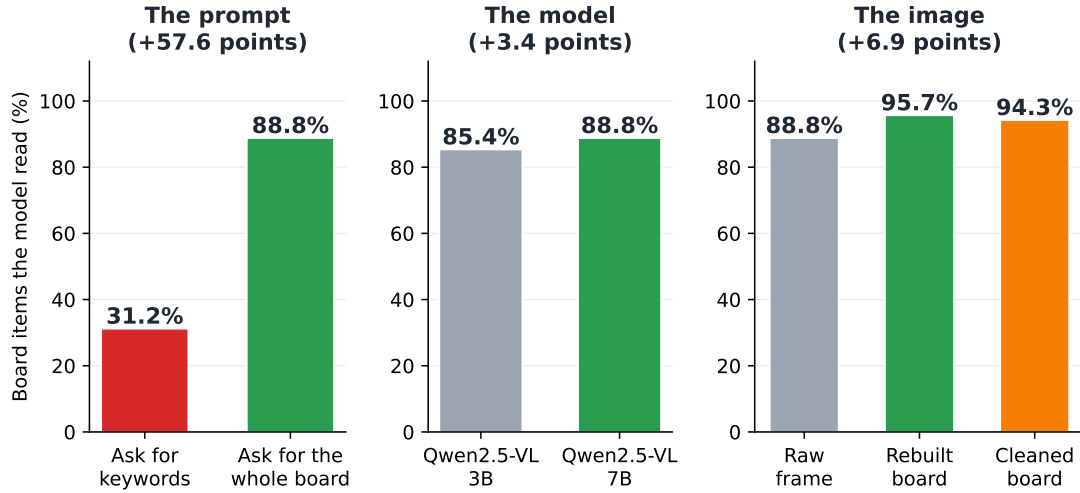
Table 5.7: Board-content recall, same model, same images, only the prompt changes.

Boards	Keyword prompt	Full transcription	Keyword to transcription
10 boards, 194 items	41.2%	96.9%	better on 9, worse on 0, sign $p = 0.0039$
35 boards, 349 items	31.2%	88.8%	better on 34, worse on 0, sign $p = 1.2e-10$
A third lecturer, 10 boards, 87 items	not run	89.7%	–

Figure 5.4 shows the three comparisons side by side. Broken down by kind over the 35 boards, the change is from 0 of 67 numbers to 65 of 67, from 43 of 47 names to 45 of 47, from 14 of 111 code fragments to 90 of 111, from 47 of 94 terms to 89 of 94, and from 5 of 30 phrases to 21 of 30. The keyword prompt found not one number on any board. This is a result about prompting and not about model capability.

The prompt was chosen on the same boards this reports, and there was no held-out set when that choice was made. To give a figure that selection cannot explain, the boards were split by lecture using a rule fixed before any per-half number was examined, with odd numbered lectures as the development half and even numbered ones as the test half. The split is by lecture and not by board because boards within a lecture share a lecturer, a topic, a camera and a whiteboard. On the development

Reading the whiteboard: 35 boards of lectures 1-9, 349 hand-verified items



One change at a time, same boards, same answer keys. What you ask for is worth 17x the model size and 8x the image processing. The cleaned board, which looks best to a person, reads slightly worse than the reconstruction it came from.

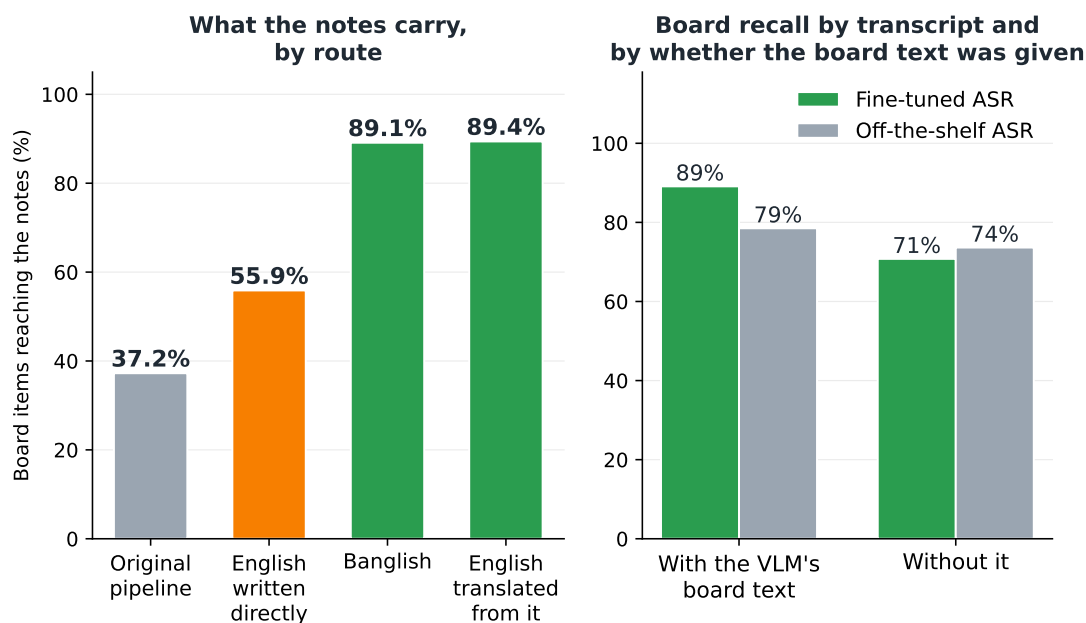
Figure 5.4: Board-content recall by item kind, keyword prompt against full transcription, over 35 boards and 349 items. The numbers written on the board go from none found to almost all found.

half the comparison is 34.1 to 93.1 per cent, better on 17 of 17 boards. On the test half, which played no part in the choice, it is 28.4 to 84.7 per cent, better on 17 of 18 boards and worse on none. This is a split applied after the original comparison rather than a protocol registered before it, so it does not undo the original selection; it answers the narrower question of whether the conclusion survives on boards that played no part in choosing the prompt.

The notes were built for all 13 lectures that have hand-verified board records, in both languages, always from a transcript produced by an adapter that never heard that lecture’s lecturer.

Table 5.8: Board-content recall of the generated notes, 35 boards of the first nine lectures, 349 items.

Notes	Recall	Against the original pipeline
Original pipeline	37.2%	–
Annotated notes, Banglish	89.1%	better on 32 boards, worse on 0, sign $p = 4.7e-10$
Annotated notes, English written directly	55.9%	better on 29, worse on 4, sign $p = 1.1e-05$
Annotated notes, English translated from the Banglish	89.4%	–
Pasting the raw board transcription into the notes	88.0%	better on 30, worse on 1



35 boards, 349 hand-verified items. Right: by board the split is 7 better / 6 worse with board text and 10 / 9 without, so the item totals overstate it. Board recall cannot see whether the surrounding explanation is right - only people can.

Figure 5.5: How much of the board reaches the notes, for the original pipeline and for the annotated notes in each language and route.

Figure 5.5 shows how much of the board reaches the notes by route. Over all 13 lectures and 436 items the Banglish notes reach 87.2 per cent and the translated English notes 87.4. On the third lecturer's ten boards, for which no earlier notes exist, the Banglish notes reach 79.3 per cent.

The comparison that matters most is with the last row of Table 5.8. Pasting the vision-language model's raw board transcription into the notes reaches 88.0 per cent and the annotated notes reach 89.1, which a paired test cannot separate at 14 boards better, 8 worse and $p = 0.29$. The paste reaches its number by being a paste, while the annotated notes reach the same number as a readable document with numbered boxes, quotations and a step by step explanation.

Across the 13 Banglish files the checker kept 31 quotations that appear word for word in the transcript and deleted 8 that do not, a rejection rate of 21 per cent. The English files contain 27 quotations given as translations, and the word-for-word check does not run on them, because a translation cannot be matched against a Banglish transcript. The supportable claim is therefore specific. Quotations in the Banglish notes are verified word for word, and quotations in the English notes are model translations and are not verified. One reference to a box that does not exist survived across the 13 Banglish files, and it is counted in the output instead of hidden.

5.1.7 Asking readers which note helps more

Board-content recall measures how much of the board survives into a note. It cannot say whether the note is pleasant to read or whether a student would rather study from it, and those are questions about people. A small reader study was therefore run.

Twenty readers were shown two sets of notes for the same lecture, the digital logic class of BanglaASR11, and asked four questions about them. One set was produced by the original pipeline, which writes English prose from the transcript alone. The other was produced by the system described in this research, which adds the rebuilt board images, the numbered box references and the checked quotations. Its explanatory prose is largely in English as well, and the lecturer is quoted in Banglish, so the comparison is not primarily one of language. The two pages were shown as Note A and Note B with every mention of a model or a method removed from them, so a reader could not tell which system produced which page. Table 5.9 gives the result.

Table 5.9: Reader study. Twenty readers, one lecture, the two notes shown blind. The test is a two-sided exact binomial test on the readers who expressed a preference, which is the sign test for paired preferences; readers who answered that the two were about the same are excluded rather than split.

Question	This research	Original	Same	p
Which set do you prefer overall	15	4	1	0.019
Which is easier to read and understand	10	4	6	0.180
Which has the better layout	18	2	0	0.0004
Which explains the concepts more clearly	10	2	8	0.039

Readers preferred the notes produced by this system overall, 15 of the 19 who expressed a preference, and the margin on layout is the largest in the study at 18 of 20. The question about explaining concepts also favours this system, 10 of 12. The question about ease of reading does not reach significance, 10 of 14 at $p = 0.18$, and it is the one result of the four that should not be claimed.

Five things limit what this study can support and all of them matter. It is a pilot of twenty readers on one lecture, recruited from people available to the team rather than sampled. No correction for multiple comparisons has been applied, and under a Bonferroni correction across the four questions only the layout result would remain, so the overall preference and the concept question should be read as suggestive rather than settled. The two notes differ in several ways at once, since the new note has board images, box references, checked quotations, Banglish quotations of the lecturer and three times the length, so the study measures the pipeline as a whole and cannot say which of those changes the readers responded to. The order was not randomised and Note A was always shown first, which if anything works against the result reported here, because the first item seen usually has the advantage. And no reader was asked to study from either note and then answer questions about the lecture, which is the test that would measure learning rather than preference.

5.2 Analysis of Design Solutions

This section analyses the design choices against the alternatives. Four of the measurements below are negative results.

5.2.1 Three approaches that did not work

An English recogniser and a Bengali recogniser were run on the same nine lectures and their outputs merged. The merged output scores 73.9 per cent on the term measure against the baseline’s 73.2, a difference of 0.7 percentage points, with an exact paired permutation test at $p = 0.32$ and a Wilcoxon signed-rank test at $p = 0.36$. Four of the nine lectures get worse, and the word error rate of the merged output is 146.8 per cent against the baseline’s 82.6. Fusion does not fix Banglish recognition, which is what motivated fine-tuning instead.

A sweep of the decoder bias strength from 0 to 2.0 on one lecture gave term recall of 8.8 per cent with the bias off and 4.1 per cent at every non-zero setting, with the transcript truncated from 279 to 149 characters. Every non-zero strength produced byte-identical output. The shape of that result matters more than its size. This is not a tuning problem with a better setting waiting to be found, because the mechanism fails as soon as it engages. The recorded output gives the optimal bias as zero, so the experiment concluded against itself.

Where the lecturer stands might indicate what the lecture is currently about, which would give a free alignment between speech and board regions. The test needs no manual labels: if standing in front of a region means working on it, that region should gain more ink after the lecturer leaves than the regions he was not standing in front of. Over 191 occlusion episodes in nine lectures the occluded region gained more ink in 101 cases and an unoccluded region in 81, with a Wilcoxon test at $p = 0.054$ and an exact sign test at $p = 0.159$, which does not support the claim. The most likely reason is that the test measures writing, while a lecturer standing in front of a region is often explaining it and adding no ink at all, which is exactly the case where the signal would have been most useful.

5.2.2 Ablations on the vision stage

The clean-up makes every board a clean white image and was judged by eye to be better. Read by the same vision-language model with the same prompt against the same records, it is slightly worse. Over the 35 boards of the first nine lectures the mosaic scores 334 of 349 items and the cleaned board 329, and over the third lecturer’s ten boards 74 of 87 against 71. Combined over 45 boards that is 93.6 per cent against 91.7, better on 1 board and worse on 7, with a sign test at $p = 0.0703$.

The difference is not significant either way, so the statement supported by the data is that there is no measured difference, with the point estimate against the clean-up. Reading the changes item by item explains the shape. The one gain is on the board the clean-up was built for, where an end-of-era diagram that the old mask lost becomes readable and the board rises from 57.1 to 71.4 per cent. The losses are all single fine details of the kind a clean-up erases, being long code strings, a written

out sentence and two decimal numbers. There is also very little room to win, since the mosaic is already at 95.7 per cent on the first nine lectures and 37 of the 45 boards score identically on both.

The cleaned boards are therefore kept for how they look in a note rather than because they help the model read, no claim is made that the rebuild improved board reading, and the headline board figures are reported on the mosaic boards.

Two further claims the design invites were tested in the same way, by changing one thing and rescoreing. The first is that the result needs a large vision-language model. Board reading was run again with Qwen2.5-VL-3B, changing nothing but the model. Over the 35 boards of the first nine lectures the 3B model scores 85.4 per cent against the 7B's 88.8, and over the third lecturer's boards 77.0 against 89.7. Combined over 45 boards and 436 items that is 83.7 per cent against 89.0, with the 7B better on 16 boards and worse on 6 and a sign test at $p = 0.053$.

Changing the prompt is worth 57.6 percentage points, from 31.2 to 88.8. Changing the model from 3B to 7B is worth 5.3 points, and that difference is not significant by the sign test on either set on its own. A 3B model at 85.4 per cent already carries most of the board. The 7B is the better model and remains the one reported, but model scale is not what made board reading work, which agrees with the published finding that large multimodal models are strongly sensitive to how the instruction is written [53]. The 32 and 72 billion parameter versions were not run: their weights are 68.3 GB and 146.8 GB, the machine had 62 GB of free disk at the time, and both would have needed quantising to fit in 32 GB of video memory, which would confound model size with quantisation loss. This is therefore a comparison of a 3B model against a 7B model and not a scale study.

The second claim is that the reconstruction improves reading, and the measurement does not support it. On the 35 boards of the first nine lectures, recall rises from 88.8 per cent on a raw frame to 95.7 on the rebuilt board, better on 8 boards and worse on 3, with a sign test at $p = 0.23$ and a Wilcoxon test at $p = 0.056$. On the third lecturer's boards it moves the other way, from 89.7 to 85.1. This is a trend on one set and absent on the other, so the prompt is what makes the model read the board and the reconstruction is what makes the board usable in a note. Two limits of the reconstruction were exposed by the hand check. Glare from the ceiling lights sits in every frame, so no choice of frame removes it, and content visible in only one frame can be lost, which happened to a binary number that was fully written at one moment, covered ten seconds later and wiped ten seconds after that.

5.2.3 Does fine-tuning the speech model improve the notes?

This question joins the two halves of the research, and the answer is no. The reason needs explaining. Table 5.10 counts the items and Table 5.11 counts the boards. Four sets of notes were built for the 13 scored lectures, Banglish only, with everything held constant except whether the transcript came from the fine-tuned or the off-the-shelf model, and whether the board text was supplied at all.

None of the four comparisons is significant. Counted by board, the two speech conditions split 7 to 6 with board text and 10 to 9 without it, which is as close to a coin toss as this data can produce. The plus 10.6 points in the item totals is worth

Table 5.10: Board-content recall for the two-by-two, 35 boards, 349 items.

	With board text	Without board text
Fine-tuned transcript	89.1% (311 items)	70.8% (247)
Off-the-shelf transcript	78.5% (274)	73.6% (257)

Table 5.11: The same four cells counted by board, which is the test the conclusion rests on.

Comparison	Item change	Better / worse	Sign	Wilcoxon
With board text, base to fine-tuned	+10.6 pp	7 / 6	p = 1.0	p = 0.81
Without board text, base to fine-tuned	-2.8 pp	10 / 9	p = 1.0	p = 0.72
Fine-tuned, no board text to board text	+18.3 pp	13 / 7	p = 0.26	p = 0.16
Off-the-shelf, no board text to board text	+4.9 pp	10 / 4	p = 0.18	p = 0.052

taking apart, because quoting it as the benefit of fine-tuning would be wrong. Of the 37 items gained, two boards of one lecture give 50 between them, and across the other 31 boards the fine-tuned transcript is 13 items worse.

The reason the better transcript loses is mechanical. Scoring the two transcripts directly against the same records, before any notes are written, the off-the-shelf transcript with 67.7 per cent character error contains 106 of the 349 board items and the fine-tuned transcript with 15.8 per cent contains 91, better on 1 board and worse on 9 with a sign test at $p = 0.021$. Off-the-shelf Whisper translates into English, and the records are largely English technical strings such as NAND gate, MySQL and lines of SQL, so its output is unusable as a transcript while still emitting those exact strings. The fine-tuned model writes what the lecturer said, in Banglish, and *amra NAND gate dekhbo* does not string-match an English item. This is the same trap as the term measure discussed in Section 5.4.2.

The null result says that the fine-tuned transcript changes almost nothing that board-content recall can see, and that measure asks only whether facts written on the board reach the notes. It does not say the transcript is irrelevant to note quality. The measure is blind by construction to whether the explanation around those facts is correct, readable or even in the right language, and the two transcripts differ enormously as transcripts. An English page built from a transcript with 67 per cent character error will contain sentences that are simply wrong, and nothing here measures that. Testing it properly needs readers. The study in Section 5.1.7 asked twenty of them which note helps more, but it compared the whole pipeline against the original one and not one transcript against another, so it does not settle this particular question.

The two routes to the English note bear on the same question, and their comparison shows the same measurement effect from the other side. Writing the English note directly from the board and the transcript gives 55.9 per cent recall, while writing

the section in Banglish first and translating it gives 89.4, almost exactly the Banglish original’s 89.1. That supports the reading above, in which the deficit of the direct route is the model paraphrasing the board’s strings into English prose rather than the note carrying less of the lecture.

The caveat is large and is stated with the number every time. Of the 130 items gained, 113 come from four dense boards carrying 39 to 42 items each, and across the other 31 boards the two routes are level. Counted by board the comparison is 10 better, 7 worse and 18 tied, with a sign test at $p = 0.63$, and there are real regressions, with two boards falling from 6 of 7 items to 2 of 7. The translated route is therefore much better on dense, number-heavy boards and indistinguishable from the direct route on ordinary ones.

5.3 Final Design Adjustments

Evaluation changed the design in several places. What follows is what changed, what evidence forced it, and what is still wrong.

5.3.1 Changes made in response to a measurement

The loop safeguard was adopted after the small model fell into repetition loops on 58 of 184 clips under plain greedy decoding, and it was applied identically to both models. With the large model the fine-tuned system loops on one clip in 177, so the safeguard changes nothing and the reported figures are the plain ones. Both facts are reported, including that the safeguard was adopted after the greedy result had been seen.

The person mask was replaced. The original took whatever differed from the usual appearance of the board, which failed when the lecturer stood still and again when the camera’s automatic exposure darkened a frame. A pretrained segmentation network with a shadow mask replaced it and recovers writing the old mask lost. Frames are also extracted more densely for a lecturer who does not move, as measured in Section 5.1.5.

Each version is now asked for in a single language. The first design followed a mock-up in which an English note quoted the lecturer in Banglish with a translation beside it, and reading the first real notes showed that an English-medium reader cannot use a Banglish quotation. That narrows what can be claimed. The word-for-word check runs on the Banglish notes and cannot run on the English ones. The instruction is only partly obeyed, and this should be read as a limitation of the generator rather than a property of the output: in the Banglish notes the quotations are in Banglish but much of the explanatory prose around them is written in English anyway. The English note is produced by translating the Banglish note, selected on the measurement in Section 5.2.3 with the caveat about the four dense boards.

Four defects found in the real runs are handled in code and are listed here because each one changes what the output contains. A loop guard collapses repeated identical lines in the board reader, after one board produced the same line 512 times for 42,037 characters, which matters because that text is pasted into the note prompt.

A salvage parser recovers the boxes of a board whose code content made the model return broken JSON, which would otherwise lose every box on that board. A placeholder strip removes the prompt’s own example text when the model copies it into the notes as though it were a quotation, a failure the quote checker could not catch because it looks for quotations absent from the transcript and this is not a quotation at all. A label repair maps the section labels the translation left in Banglish and the summary lines it returned in capital letters. The page builder also normalises the label on the Background box at render time, after the model wrote that heading four different ways and 11 of 21 boxes in one rebuild carried no disclaimer, which defeats the purpose of the box.

5.3.2 What is still wrong

Four defects are known, counted and not fixed. The first is that the box finder treats board furniture as content. One room’s whiteboard carries a manufacturer’s sticker, and because it is dark and separate the box finder puts a box around it on every board of that lecture. It is cosmetic, costs nothing in any measurement, and would be fixed by ignoring a region that appears unchanged across every era of a lecture.

A second defect is that some quotations cannot be translated and are dropped. On the thirteen scored lectures the translation route kept 15 quotations and dropped 19, and over all 42 lectures that have notes it kept 60 and dropped 49. A dropped quotation costs the page a lecturer’s voice. It is counted in the output file of every page, and it is the largest remaining defect in the English version.

A third is that a few references to boxes that do not exist survive. Across the 13 scored Banglish files the box reference check removed all but one, and over all 42 lectures ten survive in each language.

The last is that four era boundaries fall in the middle of an erase, as described in Section 5.1.5. The fix is to hold the era open until the wipe finishes.

One further defect is not a software fault and cannot be fixed by a checker. In the notes for one digital logic lecture the model wrote that there are two types of universal gates, which are XOR gates, where the lecturer and the board both say NAND and NOR. Board-content recall cannot catch this, because it measures whether board items appear in the notes and not whether the prose around them is true. Nothing in this research measures the factual correctness of the generated explanation. The reader study in Section 5.1.7 asked which note readers preferred, not whether its statements are true, so that gap is still open, and an example of the error is more useful to a reader than a claim that it does not happen.

5.4 Statistical Analysis

5.4.1 Tests, medians and the direction of a difference

Every comparison is paired, because both systems are run on exactly the same clips or boards, and two tests are applied to each: the Wilcoxon signed-rank test, which ranks the differences by size [26], and the exact sign test, which counts only how

many items improved. Both are reported wherever they disagree, as in the two-by-two of Section 5.2.3, where one comparison sits at Wilcoxon $p = 0.052$ and sign $p = 0.18$ and neither is treated as significant.

Two reporting rules were adopted after being caught out by their absence. Medians are reported because both models sometimes fall into a repetition loop and emit far more words than were spoken, which pushes a mean error rate above 100 per cent; on one early comparison the same data gave $p = 0.76$ on means and $p = 2.19\text{e-}05$ on medians. The direction of the difference is reported because a two sided p -value does not say which system won: one early adapter was significantly worse than its baseline at $z = -2.62$ and $p = 8.7\text{e-}03$, and reporting only the p -value would have made that look like a success.

Recall over record items is dominated by dense boards, because a board with 42 items contributes fourteen times as much as one with 3, and two results here look very different depending on which is counted. The translated English route gains 130 items over the direct route, of which 113 come from four boards, and counted by board it is 10 better and 7 worse at $p = 0.63$. The two-by-two gains 37 items from fine-tuning, of which 50 come from two boards while the other 31 lose 13, and counted by board it is 7 better and 6 worse at $p = 1.0$. Both countings are reported wherever they differ and the by-board count is the one the conclusion rests on.

No correction for multiple comparisons has been applied, which is a limitation. It matters least for the speech result, where the p -values would survive any correction, and most for the marginal results, which is why every marginal result here is described as a trend or as not significant.

5.4.2 A metric that moves the wrong way

The measure inherited from the earlier phase of this project is a term based F1 over a fixed list of 82 English words. Two independent training runs on identical data, with the same recipe and seed and differing only in the order the manifest listed the clips, gave the results in Table 5.12.

Table 5.12: Two training runs on identical data, showing which measures replicate.

Measure	Run A	Run B	Spread
Word error rate, median	81.8%	77.1%	4.7 pp
Character error rate, median	60.7%	57.4%	3.3 pp
Term recall	88.8%	88.3%	0.5 pp
Term precision	66.7%	55.3%	11.4 pp
Term F1	76.1%	68.0%	8.1 pp

The speech result replicates, with both runs beating the base model at p below 0.01 on word and character error rate. Term F1 does not, swinging 8.1 points on identical data, driven by an 11.4 point swing in term precision. Term precision counts how many of the 82 fixed English words the model emits, and run B transcribes better Banglish, so it emits fewer English words and its term precision falls. The measure penalises the model for doing the task correctly.

Three consequences follow and are applied throughout this research. A term F1 difference smaller than about 8 points is never claimed, which independently rules out the fusion result of 0.7 points. Word and character error rate with their rank tests are the headline. Both runs are reported.

The same effect reappears in the note evaluation of Section 5.2.3, where the worse transcript contains more of the English record items. Two independent instances of the same failure in one project is why this research treats it as a finding rather than a nuisance, and it agrees with the published work on evaluation metrics for code-switched recognition discussed in Section 2.2.4.

5.5 Comparisons and Relationships

5.5.1 Comparison with existing systems

Table 2.1 in Chapter 2 places this work beside the four nearest Bengali-English code-switched speech systems. Their published error rates cannot be compared with each other, because two of the four are trained and evaluated on synthetic speech, one targets the Bengali script and one is an undocumented public model evaluated on a random split of its own clips. Rather than leave the comparison there, five published models were run on the test clips of this research, and Section 2.2.3 reports the result on one scale.

That comparison says two things. Not one of the five produced romanised Banglish on a single clip, which is the measured form of the claim that no existing system does what is needed here. But once their Bengali output is transliterated to Roman, the strongest of them reaches 47.1 per cent character error against this work’s 15.8, which is better than off-the-shelf Whisper and so cannot be dismissed. The gap is real and it is not only a matter of alphabet, and both halves of that sentence are needed.

The evaluation design is also stricter here in three ways. Whole lectures are held out, so no part of a test recording appears in training; the references are checked word by word by a person rather than generated; and the same system is reported under two designs, one with the test lecturers heard in training and one with an entirely unseen lecturer.

For the board side the comparison is different in kind. Systems such as those of Davila and Zanibbi [44] and Kota et al. [45] extract and binarise handwriting from fixed camera lecture video and report component level recall above 94 per cent on public datasets. This work does not beat them at that task and does not attempt to, since its reconstruction is engineering assembled from known parts. What is added is the step after it. Reading the rebuilt board into text with a vision-language model, scoring that against hand-verified records, and carrying numbered regions of the board into a generated note.

5.5.2 Relationships between data, stages and the earlier phase

Two observations connect the amount of data to the result and neither is a controlled experiment. Within the final result the lecturer with the least training speech has by far the worst test lecture, at 52 per cent character error against 12 to 16 for the others. Across designs, holding out an entire lecturer and training on 80 to 114 minutes gives 50.3 per cent character error, while holding out whole lectures from known lecturers and training on 4.10 hours gives 15.8. The two differ in both the amount of data and the difficulty of the question, so neither isolates the effect of data volume. An earlier scaling curve on the smaller model, at 0.30, 0.60 and 0.93 hours, gave 84.6, 80.6 and 83.3 per cent word error, which is flat within the 4.5 point run-to-run noise after the first 18 minutes. A curve on the current corpus was planned and cancelled when the corpus closed.

The clearest relationship between the stages is that the factual content of a note comes from the vision side and not the speech side. Adding the board text moves note recall by 18.3 points with the fine-tuned transcript and 4.9 points with the off-the-shelf one, while changing the transcript with board text present moves nothing a by-board test can detect. This is a statement about what board-content recall measures rather than a defect of the speech work, and it is the reason the speech result is reported on its own terms as a transcription result.

The earlier phase of the project reported a fusion improvement of 5.7 percentage points with $p = 0.003$ and a large effect size, an ablation table, a bias sweep and a set of per-video statistics, none of which was produced by any code in the project. They were re-derived where a derivation was possible and retired where it was not. The measured fusion effect is 0.7 points at $p = 0.32$, two of the ablation rows described modes the pipeline does not have, and a failure-mode breakdown of five neat percentages summing to 100 had no analysis behind it and was removed instead of reconstructed. This is recorded because it is the reason for the project rule that a number needs a command, and because a reader comparing this report with the earlier one is entitled to know which numbers changed and why.

5.6 Discussions

5.6.1 What the results mean

The speech result is the strongest part of the research. A small adapter on a large multilingual model, trained on 4.10 hours of hand-checked code-mixed classroom speech, cuts the character error rate from 67.7 to about 16 per cent on lectures the model has never heard, with two seeds agreeing and almost every clip improving. In practical terms the output changes from something a transcriber cannot use into a draft a transcriber can correct.

The board result is a result about prompting rather than about model capability, since the same model on the same images moves from 31.2 to 88.8 per cent recall purely on how it is asked, and a model less than half the size reaches 85.4 per cent. Anyone building a system of this kind should spend effort on the prompt before

spending it on a bigger model. The note result is that board content can be made to reach the page. The annotated notes carry 89.1 per cent of the board items against 37.2 for a conventional summarisation prompt, while remaining a readable document and not a paste of extracted text.

The negative results matter as much. Fusion, decoder bias and occlusion as a pointer were all built and all measured, and none worked; the display clean-up looks better and reads slightly worse; and fine-tuning the speech model does not move note recall. Reporting these is what makes the positive results credible.

5.6.2 Implications

For code-mixed speech, the data is the bottleneck and the method is not exotic. A standard adapter on a standard model gives a large gain, and what took the effort was five hours of carefully checked transcription and a split design that does not leak.

For evaluation, measures assembled out of English strings systematically reward a system that translates code-mixed speech and punish one that transcribes it. It happened twice here in two unrelated measures, and it is consistent with the published work on code-switched evaluation reviewed in Section 2.2.4. Anyone evaluating a code-mixed system on an English-derived metric should check for this before believing the result.

For classroom systems, the whiteboard is worth the trouble. Most of the facts a student needs from these lectures are on the board and not in the speech, and a pipeline that reads the board recovers them.

5.6.3 Limitations and threats to validity

The limitations are listed together rather than spread through the chapter.

1. The human evaluation is a pilot. Twenty readers on one lecture preferred these notes to the original pipeline's, but the two pages differ in several ways at once, the order was not randomised and no reader was asked to learn from either page. Nothing here measures whether the generated explanation is true, and a factual error in the prose has already been found by reading.
2. Three lecturers, one university, one language pair. Nothing here shows that the result transfers to another institution, subject area or code-mixed pair.
3. 5.15 hours of transcribed speech, which is small by the standards of speech recognition, and the relationship between hours and accuracy at this scale is not measured.
4. Training is unstable at the selected learning rate, since one seed of the pre-registered configuration diverged completely, and the stable configuration reported beside it was chosen after that was seen.

5. The board-reading prompt was chosen on the boards it is reported on. A held-out half was checked afterwards and supports the conclusion without undoing the selection.
6. Box finding is not evaluated, because no layout ground truth exists. It works well on sparse boards and coarsely on dense ones, and it draws a box around a sticker. The vision-language model can also locate the regions itself rather than being given them, and the box finder has an option for that, but every result reported here uses the boxes measured from the ink and the alternative was not run.
7. Quotations in the English notes are not verified, since they are translations and the automatic check cannot apply to them.
8. One wav2vec 2.0 Bengali model in the baseline comparison was not run, only its output alphabet inspected, so Section 2.2.3 covers five published models and not every candidate.
9. No correction for multiple comparisons, which is why marginal results are described as trends.
10. The system is for recorded lectures and nothing here runs in real time.

The main threat to the speech result would be leakage between training and test. Whole lecture splits, the voice check on every recording and the frozen record of which lectures were on each side of every split guard against it. One leak did occur earlier in the project, it was found by the voice check, and the affected claim was retired.

The main threat to the vision results is that the board records were drafted by reading the rebuilt boards, which could favour the reconstruction over a raw frame. The records were then checked by a person against the board images without seeing any model output, and the comparison between the reconstruction and the raw frame is reported as a trend rather than a result, so the conclusion does not depend on it.

The main threat to the note results is that board-content recall is a narrow measure that a system could satisfy by pasting the board text into the page. That is exactly what the variant at 88.0 per cent does, which is why the annotated notes are compared against it rather than only against the original pipeline.

Chapter 6

Conclusion

6.1 Summary of Findings

This research set out to turn a recorded Bengali-English code-mixed whiteboard lecture into a lecture note a student can use. The system was built as four stages and each stage was measured separately.

Off-the-shelf speech recognition does not work on this speech, and the failure is specific. On six held out lectures Whisper large-v3-turbo produces a median character error rate of 67.7 per cent and a median word error rate of 93.9, translating the speech into English instead of transcribing it and falling into repetition loops on thirteen of the 177 test clips. A small adapter on the same model corrects most of that. Trained on 4.10 hours of hand-checked Banglish from 22 lectures, LoRA adapters bring the character error rate down to 15.8 and 16.0 per cent in two random seeds and the word error rate to 42.5 and 41.7, better on 164 and 167 of 177 clips and needing no decoding safeguard. The gain also holds for a lecturer the model has never heard, at a lower level, cutting the character error rate from 72.8 to 50.3 per cent in all six runs. Training at the selected settings is not stable, however, since one of two seeds at the chosen learning rate collapsed into repetition and produced output worse than doing nothing.

The whiteboard can be rebuilt from the video without inventing anything, and the median board has 97.7 per cent of its tiles fully clear. A human check of 98 boards found 82 complete, with writing genuinely lost on 9 and the remaining 7 explained by the source video and not by the method. Reading the board is then a question of prompting rather than of model size. The same vision-language model on the same images moves from 31.2 to 88.8 per cent recall of hand-verified board content when the prompt asks for a full transcription instead of keywords, and from none of 67 numbers to 65 of 67, while a model less than half the size reaches 85.4 per cent. Board content can be made to reach the notes, which carry 89.1 per cent of the board items against 37.2 for a conventional summarisation prompt.

Several approaches did not work, and that is part of the result. Dual recogniser fusion gives 0.7 percentage points at $p = 0.32$, and biasing the decoder towards board words halves term recall at every non-zero strength. Using occlusion as a pointer is not supported, at $p = 0.054$ and $p = 0.159$. The display clean-up makes the model read boards very slightly worse, and fine-tuning the speech model does

not measurably improve board-content recall in the notes.

Measures built out of English strings mislead in this setting, and this appeared twice independently. A term based F1 over a fixed English word list swings 8.1 points between two training runs on identical data and moves against transcription quality, because a better Banglish transcript contains fewer English words. In the note evaluation the transcript with 67.7 per cent character error contains significantly more of the English record items than the one with 15.8 per cent, for the same mechanical reason.

6.2 Contributions to the Field

Stated as narrowly as the evidence allows, the contributions are the following. They are ordered by how far each one reaches beyond this dataset, because the corpus is the most visible contribution but not the most transferable one.

1. A demonstration that the prompt, rather than the model size, is what makes a vision-language model read a handwritten board. On the same 35 boards with the same model, changing only the prompt moved board-content recall from 31.2 to 88.8 per cent, while going from the 3 billion parameter model to the 7 billion parameter one moved it from 83.7 to 89.0. The prompt is worth 57.6 points and the model size 5.3, about eleven times less. The prompt was re-checked on a held-out half of the boards, where it gives 28.4 to 84.7 per cent, so the conclusion does not rest on the boards the prompt was chosen on. Nothing about this finding is specific to Banglish or to this corpus.
2. A note-quality measure that a language model cannot satisfy from prior knowledge. Board-content recall scores only facts physically written on the board, which the summarising model cannot invent from what it already knows, so it separates grounding from world knowledge. It behaves as a valid measure should: recall falls as items become more guessable. Generated lecture notes are usually evaluated by fluency or by overlap with a reference summary, and neither can tell whether the note is about the lecture that was actually given.
3. A measured demonstration that measures assembled from English strings are invalid for code-mixed speech, because they reward a model that translates and penalise one that transcribes. The inherited term measure moves in the wrong direction as transcription improves, since better Banglish emits fewer of the English words the measure counts, and it moves by 8.1 points between two runs on identical data. This was observed twice in unrelated measures within one project and it agrees with the published literature on code-switched evaluation.
4. The first measurement, as far as could be found, of what published Bengali and Bengali-English speech models actually produce on spontaneous code-mixed classroom speech. Five models were run on the same held-out clips as this work. Not one produced romanised Banglish on a single clip: four write Bengali script and the fifth translates into English. The finding survives the obvious objection, since transliterating their output to Roman and scoring

again still leaves the best of them at 47.1 per cent character error against 15.8 here.

5. A hand-checked corpus of code-mixed classroom speech in romanised Banglish. Forty-four lecture recordings totalling 8.33 hours, of which 28 lectures and 5.15 hours are transcribed word by word against the audio according to a written standard, with 609 timed segments and about 39,100 words. No comparable corpus of spontaneous Bengali-English classroom speech transcribed in the Roman alphabet could be found, and the nearest published corpora in this language pair are synthetic.
6. A measured speech recognition result for this variety under a strict evaluation design. Whole lectures are held out, references are human, the settings come from a procedure written down before any run, and two seeds are reported for every configuration including the one that diverged. Two designs are reported and labelled, one with the test lecturers heard in training and one with an entirely unseen lecturer.
7. A whiteboard reconstruction that generates nothing, read into text by a vision-language model and scored against hand-verified records. The reconstruction itself is engineering assembled from known parts, and what is new is the combination with a vision-language reader and the measurement of it, including 45 boards and 436 items verified by hand and 98 boards checked for missing writing.
8. A note generator whose claims can be checked. Board content is supplied as text instead of being recalled, non-lecture material is confined to a labelled box, quotations are verified word for word against the transcript and deleted when they fail, and references to numbered regions of the board are checked against the regions that exist. Each of these is counted in the output file of every page.
9. A set of measured negative results, since fusion, decoder bias, occlusion as a pointer and the display clean-up were all built, all measured and all reported as they came out.

6.3 Recommendations for Future Work

The reader study reported in Section 5.1.7 is a pilot and the obvious thing to extend. Twenty readers on one lecture preferred the notes this system produces, but the comparison moved several things at once and no reader was asked to learn from either note. A fuller study would randomise the order, cover several lectures and lecturers, separate the changes so that the board images, the box references and the language can be judged one at a time, and above all ask readers to answer questions about the lecture after studying a note, which measures learning rather than preference. A second arm should compare notes built from the off-the-shelf transcript with notes built from the fine-tuned one, which is the only way to test the part of the speech improvement that board-content recall cannot see.

Measuring whether the generated prose is true is still untouched. A factual error has already been found by reading, and nothing in this research counts how often that happens. The check needs a reader who knows the subject going through the notes statement by statement against the recording.

One smaller piece of evidence would also strengthen the report. Timing transcribers as they correct real model drafts would replace the projection in Section 3.8 with a measurement, and it is the number a department would actually want.

On the speech side, more data from under-represented lecturers is the clearest next step, since the lecturer with 0.7 hours in training has a test lecture at 52 per cent character error while the others reach 12 to 16, and collecting more from that lecturer would test whether the cause is the data or the speaker. A scaling curve on the current corpus, planned and cancelled when the corpus closed, would answer how much more transcription is worth buying. The divergence at the selected learning rate is reproducible enough to study, and warmup, gradient clipping or a lower scaling factor are the obvious things to try; a configuration that never diverges would be worth more than a small gain in error rate. Reporting the spelling tolerant variants of Section 5.1.3 as standard, and applying the same idea to the note measure, would reduce the measurement problem described throughout this research.

On the vision and note side, four changes have known causes. Holding an era open until a wipe finishes would fix the four boards cut in the middle of an erase. Ignoring a region that appears unchanged in every era of a lecture would stop the box finder putting a numbered box around a sticker. Evaluating the box finding would attach a number to the one part of the pipeline that has none, and it needs layout ground truth that does not yet exist. Notes in the Bengali script were dropped from this research deliberately, since they would be produced by translation alone, and they are straightforward to add if a reader wants them. A larger notes model was intended for comparison and dropped for disk space; it would change the writing rather than the facts, since it reads the extracted board text and never the board.

The system does what it was built to do, and this report states how well. A student can put in a recorded Banglish whiteboard lecture and get out a note containing the lecturer's words and the lecturer's board, with the board drawn from real pixels and the quotations checked against the transcript. The speech model is good enough to be a draft and not good enough to be trusted unchecked, the board reading is strong and the prompt is the reason, and whether the notes are genuinely useful to a student is the one question this research cannot answer.

Bibliography

- [1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023, pp. 28 492–28 518.
- [2] E. J. Hu et al., “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [3] S. Bai et al., “Qwen2.5-VL technical report,” *arXiv preprint arXiv:2502.13923*, 2025.
- [4] Qwen Team, “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2025.
- [5] J. Yang, H. Zhang, F. Li, X. Zou, C. Li, and J. Gao, “Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V,” *arXiv preprint arXiv:2310.11441*, 2023.
- [6] M. T. Agro, A. Kulkarni, K. Kadaoui, Z. Talat, and H. Aldarmaki, “Code-switching in end-to-end automatic speech recognition: A systematic literature review,” *arXiv preprint arXiv:2507.07741*, 2025.
- [7] M. Fahim et al., “BanglaTLit: A benchmark dataset for back-transliteration of romanized Bangla,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, Association for Computational Linguistics, 2024, pp. 14 656–14 672.
- [8] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, “Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks,” in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006.
- [9] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] A. Gulati et al., “Conformer: Convolution-augmented transformer for speech recognition,” in *Proc. Interspeech 2020*, 2020.
- [11] A. Baeovski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [12] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2021.

- [13] S. Chen et al., “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, 2022.
- [14] A. Mohamed et al., “Self-supervised speech representation learning: A review,” *IEEE Journal of Selected Topics in Signal Processing*, 2022.
- [15] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, “Unsupervised cross-lingual representation learning for speech recognition,” in *Proc. Interspeech 2021*, 2021.
- [16] A. Babu et al., “XLS-r: Self-supervised cross-lingual speech representation learning at scale,” in *Proc. Interspeech 2022*, 2022.
- [17] D. S. Park et al., “SpecAugment: A simple data augmentation method for automatic speech recognition,” in *Proc. Interspeech 2019*, 2019.
- [18] M. Bain, J. Huh, T. Han, and A. Zisserman, “WhisperX: Time-accurate speech transcription of long-form audio,” in *Proc. Interspeech 2023*, 2023.
- [19] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, 2020, pp. 1906–1919.
- [20] N. Houlsby et al., “Parameter-efficient transfer learning for NLP,” in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019.
- [21] X. L. Li and P. Liang, “Prefix-tuning: Optimizing continuous prompts for generation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.
- [22] T. Detrmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized llms,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [23] H. Yang, Y. Peng, H. Huang, and S. Li, “Adapting Whisper for parameter-efficient code-switching speech recognition via soft prompt tuning,” in *Proc. Interspeech 2025*, 2025, pp. 5203–5207. DOI: 10.21437/Interspeech.2025-2549
- [24] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [25] T. Wolf et al., “Huggingface’s transformers: State-of-the-art natural language processing,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, 2020.
- [26] F. Wilcoxon, “Individual comparisons by ranking methods,” *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
- [27] F. R. Rakib et al., “OOD-Speech: A large bengali speech recognition dataset for out-of-distribution benchmarking,” *arXiv preprint arXiv:2305.09688*, 2023.
- [28] R. Ardila et al., “Common Voice: A massively-multilingual speech corpus,” in *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, 2020.

- [29] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, and R. Collobert, “MLS: A large-scale multilingual dataset for speech research,” in *Proc. Interspeech 2020*, 2020.
- [30] T. Javed, K. S. Bhogale, A. Raman, A. Kunchukuttan, P. Kumar, and M. M. Khapra, “Indicsuperb: A speech processing universal performance benchmark for indian languages,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.
- [31] H. A. Z. S. Shahgir, K. S. Sayeed, and T. A. Zaman, “Applying wav2vec2 for speech recognition on Bengali common voices dataset,” *arXiv preprint arXiv:2209.06581*, 2022.
- [32] B. Roark et al., “Processing south asian languages written in the latin script: The Dakshina dataset,” in *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, 2020.
- [33] Y. Madhani et al., “Aksharantar: Open Indic-language transliteration datasets and models for the next billion users,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- [34] S. Sitaram, K. R. Chandu, S. K. Rallabandi, and A. W. Black, “A survey of code-switched speech and language processing,” *arXiv preprint arXiv:1904.00784*, 2019.
- [35] D.-C. Lyu, T.-P. Tan, E. S. Chng, and H. Li, “SEAME: A Mandarin-English code-switching speech corpus in south-east Asia,” in *Proc. Interspeech 2010*, 2010.
- [36] G. I. Winata, S. Cahyawijaya, Z. Liu, Z. Lin, A. Madotto, and P. Fung, “Are multilingual models effective in code-switching?” In *Proceedings of the 5th Workshop on Computational Approaches to Linguistic Code-Switching*, 2021.
- [37] H. S. Chadha, P. Shah, A. Dhuriya, N. Chhimwal, A. Gupta, and V. Raghavan, “Code switched and code mixed speech recognition for indic languages,” *arXiv preprint arXiv:2203.16578*, 2022.
- [38] P. Ghosh, “MediBeng Whisper Tiny: A fine-tuned code-switched Bengali-English translator for clinical applications,” *medRxiv*, 2025, Preprint.
- [39] A. Mitra, T. Mondal, A. Chakravarty, and S. Bandyopadhyay, “Bengali-English and Hindi-English code mixed speech data with disfluencies,” in *Proceedings of the 8th Workshop on Indian Language Data: Resources and Evaluation (WILDRE)*, ELRA Language Resources Association, 2026, pp. 39–48.
- [40] I. Hamed et al., “Benchmarking evaluation metrics for code-switching automatic speech recognition,” in *IEEE Spoken Language Technology Workshop (SLT)*, 2022.
- [41] J. Emond, B. Ramabhadran, B. Roark, P. Moreno, and M. Ma, “Transliteration based approaches to improve code-switched speech recognition performance,” in *IEEE Spoken Language Technology Workshop (SLT)*, 2018, pp. 448–455.
- [42] K. Kadaoui, M. Al Ali, H. O. Toyin, I. Mohammed, and H. Aldarmaki, “Poly-WER: A holistic evaluation framework for code-switched speech recognition,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, Association for Computational Linguistics, 2024, pp. 6144–6153.

- [43] C.-H. Yeh, G.-J. Lee, B.-C. Hong, and C.-H. Tang, “Robust handwriting extraction and lecture video summarization,” *Multimedia Tools and Applications*, vol. 75, pp. 7067–7085, 2016.
- [44] K. Davila and R. Zanibbi, “Whiteboard video summarization via spatio-temporal conflict minimization,” in *14th IAPR International Conference on Document Analysis and Recognition (ICDAR)*, IEEE, 2017, pp. 355–362.
- [45] B. U. Kota, K. Davila, A. Stone, S. Setlur, and V. Govindaraju, “Generalized framework for summarization of fixed-camera lecture videos by detecting and binarizing handwritten content,” *International Journal on Document Analysis and Recognition (IJDAR)*, vol. 22, pp. 221–233, 2019.
- [46] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, “LayoutLM: Pre-training of text and layout for document image understanding,” in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020.
- [47] G. Kim et al., “OCR-free document understanding transformer,” in *European Conference on Computer Vision (ECCV)*, 2022.
- [48] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [49] OpenAI, “GPT-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [50] Y. Ding et al., “A survey on MLLM-based visually rich document understanding: Methods, challenges, and emerging trends,” *arXiv preprint arXiv:2507.09861*, 2025.
- [51] Z. Chen et al., “M3AV: A multimodal, multigenre, and multipurpose audio-visual academic lecture dataset,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, 2024, pp. 9041–9060.
- [52] R. Zhao et al., “NoteIt: A system converting instructional videos to interactive notes through multimodal video understanding,” in *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST)*, ACM, 2025.
- [53] M. I. Ismithdeen, M. U. Khattak, and S. Khan, “Promptception: How sensitive are large multimodal models to prompts?” In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Association for Computational Linguistics, 2025.
- [54] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, “Rethinking atrous convolution for semantic image segmentation,” *arXiv preprint arXiv:1706.05587*, 2017.
- [55] E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in NLP,” in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, 2019, pp. 3645–3650.

Appendix A: The Transcription Standard

Every reference transcript in this research was produced against the written standard summarised here, version 1.0. The full document was given to the transcription team before any file was written, and a validation script checks every file against it.

A.1 Segments and file format

No segment may be longer than 30 seconds, because that is the length of audio the speech model sees at once. Anything longer is cut by software that does not know where sentences end, which produces audio that does not match its text. The target is 10 to 25 seconds per segment, and a break is always taken at a natural pause. The nine transcripts produced before the standard existed do not follow this rule: their segments run to a median of 79 seconds and up to 225, and 98 per cent are over 30 seconds, so they are subdivided at sentence ends by the data preparation step.

Each video has one UTF-8 text file, named `BanglaASR<N>_ground.truth.txt`, with a header of comment lines and then one line per segment.

```
# BanglaASR12 Ground Truth Transcription
# Topic: Python Functions
# Speaker ID: B
# Video duration: 14:22
# Style guide version: 1.0
# =====
[0:00-0:16] Assalamualaikum shobai. Aj amra dekhbo je kivabe python e
           function likhte hoy.
[0:16-0:33] Function mane hocche ekta code block jeta amra baar baar call
           korte pari.
```

The timestamp is exactly `[M:SS-M:SS]` with no space inside the brackets, and the end timestamp of one segment is the start of the next, so there are no gaps and no overlaps. Lines beginning with `#` are ignored by the software.

A.2 Write it exactly as spoken

Fillers are kept, every one of them, every time, including *um*, *uh*, *aaa*, *mane*, *je*, *tahole* and *thik ache*. Three reasons were given to the team. The model has to learn

them, and in the existing files fillers are about 5 per cent of tokens, at 671 out of 12,890. Removing them breaks the alignment between the text and its time window. And a reference cleaner than reality makes every error rate reported from it wrong in our favour.

The one thing removed is an abandoned restart, where the lecturer begins a sentence, stops and starts again differently; only the finished version is kept, and anything ambiguous is kept. The lecturer is never corrected, so slips, repeats and self-corrections are transcribed as they happen.

A.3 Spelling and validation

Banglish has no spelling standard, so the guide fixes one. If one transcriber writes *kore* and another *koree* for the same word, the model sees two unrelated words and learns both badly. The canonical list was derived from what the team already wrote most often. Table 6.1 gives a sample. Technical terms stay in English, and the text is ASCII only with no Bengali script.

Table 6.1: A sample of the canonical spelling list, which the validator enforces.

Meaning	Use this	Do not write
I, my	<i>ami, amar</i>	<i>amr, aami</i>
we	<i>amra</i>	<i>aamra, amraa</i>
this	<i>eta, ei</i>	<i>etaa, eee</i>
one, a	<i>ekta</i>	<i>ektaa, akta, ekhta</i>
now	<i>ekhon</i>	<i>ekhone, akhon</i>
is happening	<i>hocche</i>	<i>hoche, hochhe, hoicche</i>
is, exists	<i>ache</i>	<i>achhe, ase</i>
if	<i>jodi</i>	<i>zodi</i>
but	<i>kintu</i>	<i>kintuh</i>
then, so	<i>tahole</i>	<i>tahle</i>
good	<i>bhalo</i>	<i>valo, bhaalo</i>
plural marker	<i>gula</i>	<i>gulo</i>

A script checks every transcript against these rules and reports segments over 30 seconds, unreadable timestamps, gaps and overlaps, missing headers and spellings outside the canonical list. It can repair malformed timestamps automatically, keeping a backup of the original file. The checks that would corrupt training silently, such as a transcript with no matching video or a lecturer label that disagrees with the voice check, are a hard gate that stops a training run before it starts.

Appendix B: Prompts Used in the Pipeline

The results in Chapter 5 depend heavily on how the models are asked, so the three prompts that matter are printed here as they appear in the code.

B.1 Reading the whole board

This prompt produced the change from 31.2 to 88.8 per cent board-content recall in Section 5.1.6. The prompt it replaced asked the model for keywords visible on the board.

```
This is a photograph of a classroom whiteboard during a university lecture.
```

```
Transcribe everything written on the whiteboard, exactly as written.
```

- Copy every number, name, identifier, date and symbol exactly. Do not round, correct or reformat values.
- Write tables as Markdown tables with the same rows and columns as the board.
- Write code, SQL and program text inside ‘‘ fences, keeping line breaks.
- Write formulas as they appear. For a bar over a term write it as NOT(...), for example NOT(A+B).
- Keep diagrams short: name what they are and copy any labels on them, for example "Gate diagram: inputs A, B into a box labelled NAND, output NOT(AB)".
- If part of the board cannot be read, write [illegible]. Never guess, and never add anything that is not written on the board.
- Ignore any person in the picture. Describe only what is written.

```
Output only the transcription.
```

B.2 Naming and transcribing each numbered box

The Set-of-Mark prompt of Section 4.4.3. The boxes are drawn on the image by our own code and numbered, and the model answers per number. Boxes it names none are dropped, which is how a smudge or a sticker can be removed by the model even though the box finder created it.

```
This is a classroom whiteboard from a university lecture. Numbered coloured boxes have been drawn on it; each box surrounds one part of the writing. There are {n} boxes, numbered 1 to {n}.
```

For every box, reply with:

- "id": the box number.
- "name": 1 to 4 words saying what kind of content the box holds, for example "Title", "Definition", "Truth table", "Gate symbol", "Block diagram", "SQL query", "Code", "List", "Worked example", "Formula". If the box holds no lecture writing (a smudge, a shadow, a logo or sticker, the edge of the board), use "none".
- "text": everything written inside that box, copied exactly: every number, name and symbol. For a bar over a term write NOT(...). Write a table as Markdown table rows. Write [illegible] for what cannot be read. Never guess.

Reply with JSON only: a list with one object per box, in box order, like [{"id": 1, "name": "Title", "text": "Digital Logic Design"}]

B.3 Writing one section of the notes

These rules are shared by both languages. Rule 1 is the direct answer to the textbook recitation described in Section 1.3, and rule 2 is what makes the notes point at the numbered boxes.

RULES

1. Teach what THIS lecturer taught, with the lecturer's own examples and values. Explain it properly, for a student who missed the class: walk through each step on the board, say what it means and why the lecturer does it in that order, and spell out anything the board only shows as a symbol. What is not wanted is material from elsewhere: every fact in the explanation must come from the board or the transcript. Anything else belongs in the BACKGROUND section, never mixed in.
2. The board has numbered coloured boxes, listed below. Point the student at them, for example "Look at the orange box 3" or "the truth table in box 5". Use only the box numbers that are listed, and mention every listed box at least once.
3. Copy numbers, names, code, formulas and table values exactly as they appear in the box text. A wrong value is worse than a missing one.
4. The transcript is machine-made and has errors. Where it is garbled, rely on the board. Never invent what the lecturer said.
5. No greetings, no filler, no "in this section".
6. A table on the board is written as a Markdown table, never described row by row.
7. Plain text for formulas and symbols: $A + B$, $(A + B)'$, $\text{NOT}(A + B)$, $A \text{ XOR } B$.

The quotation instruction differs by language. The Banglish note quotes the lecturer's real words, so the quotation can be checked against the transcript character by character, and a quote that is not word for word in the transcript is deleted automatically. The English note is read by a student who cannot read Banglish, so the lecturer is quoted in English translation; a translation cannot be matched against a Banglish transcript, so the automatic checker does not run on those files and each file records that fact in its footer.

The example line in the Banglish quotation instruction, "exact words from the transcript", was itself copied into the notes as if it were a real quotation three times across 26 files. The quotation checker could not catch it, because it looks for quotations absent from the transcript and this is not a quotation at all, so a

separate step now removes the prompt's own placeholder text and counts how many it removed.